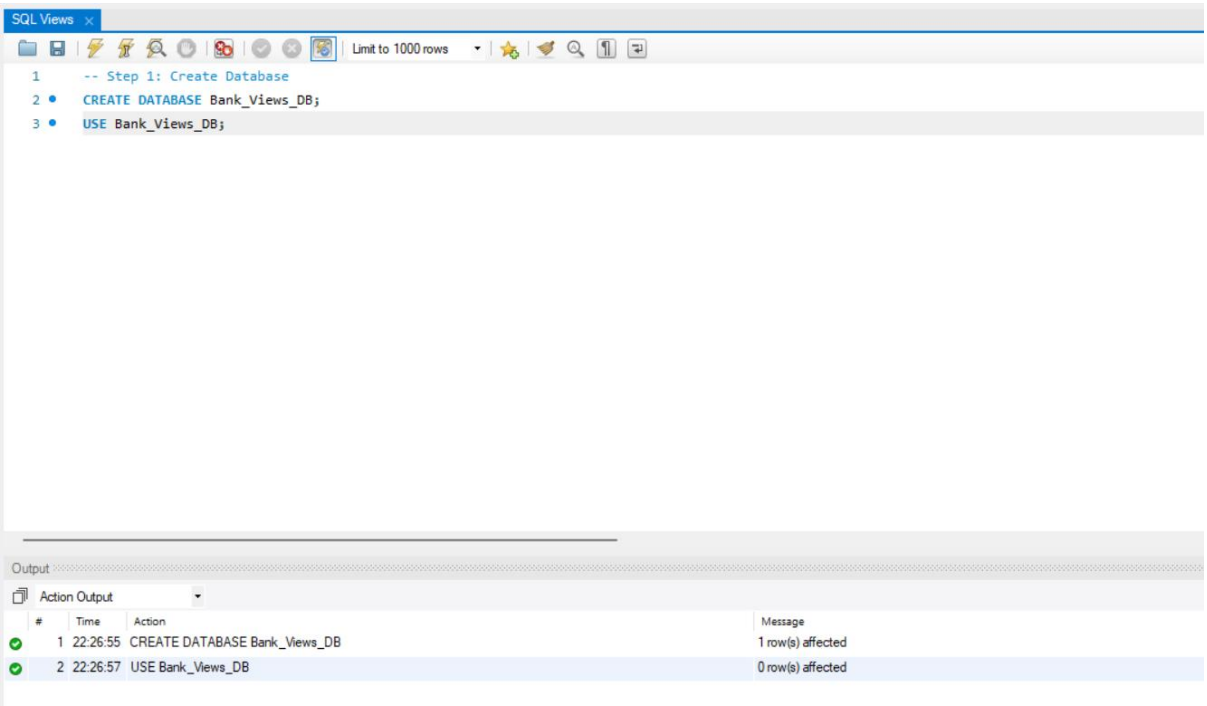
SQL VIEWS

Ili.Harshika (2520030607)

Question :

Develop a Bank Database using Customer, Account, Bank_Transaction and Loan tables. Insert sample records into all tables and implement SQL Views to display customer and account information, savings and current accounts, high/low balance accounts, branch-wise and account-type-wise summaries, customer-account and customer-loan details, transaction information, loan interest calculations, and other banking information. Demonstrate views using conditions, JOIN, aggregate functions, GROUP BY, HAVING, ORDER BY, calculated columns, UPDATE, INSERT, DELETE, view information and DROP operations.

Step 1 : Create Database and Table



SQL Views

Limit to 1000 rows

```sql
-- Step 1: Create Database
CREATE DATABASE Bank_Views_DB;
USE Bank_Views_DB;
-- Step 2: Create Customer Table
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
```

Output

Action Output

#	Time	Action	Message
1	22:26:55	CREATE DATABASE Bank_Views_DB	1 row(s) affected
2	22:26:57	USE Bank_Views_DB	0 row(s) affected
3	22:27:29	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected

SQL Views

Limit to 1000 rows

```sql
CREATE DATABASE Bank_Views_DB;
USE Bank_Views_DB;
-- Step 2: Create Customer Table
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
-- Step 3: Create Account Table
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action	Message
1	22:26:55	CREATE DATABASE Bank_Views_DB	1 row(s) affected
2	22:26:57	USE Bank_Views_DB	0 row(s) affected
3	22:27:29	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
4	22:27:59	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected

SQL Views

Limit to 1000 rows

```
);
-- Step 3: Create Account Table
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
-- Step 4: Create Bank Transaction Table
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (Account_No) REFERENCES Account(Account_No)
);
```

Output

Action Output

#	Time	Action	Message
1	22:26:55	CREATE DATABASE Bank_Views_DB	1 row(s) affected
2	22:26:57	USE Bank_Views_DB	0 row(s) affected
3	22:27:29	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
4	22:27:59	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
5	22:28:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected

SQL Views

Limit to 1000 rows

```
);
-- Step 4: Create Bank Transaction Table
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (Account_No) REFERENCES Account(Account_No)
);
-- Step 5: Create Loan Table
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action	Message
1	22:26:55	CREATE DATABASE Bank_Views_DB	1 row(s) affected
2	22:26:57	USE Bank_Views_DB	0 row(s) affected
3	22:27:29	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
4	22:27:59	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
5	22:28:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected
6	22:28:48	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), L...	0 row(s) affected

SQL Views

```
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
-- Step 6: Insert Customer Data
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101,'Harshika','9876543210','harshika@gmail.com','Hyderabad'),
(102,'Ravinder','9876543211','ravi@gmail.com','Vijayawada'),
(103,'Santhvin','9876543212','santhvin@gmail.com','Bangalore'),
(104,'Srilatha','9876543213','srilatha@gmail.com','Chennai'),
(105,'Varshika','9876543214','varshika@gmail.com','Hyderabad'),
(106,'Anirudh','9876543215','anirudh@gmail.com','Delhi'),
(107,'Goutham','9876543216','goutham@gmail.com','Mumbai'),
(108,'Zaizai','9876543217','zaizai@gmail.com','Pune'),
(109,'Manu','9876543218','manu@gmail.com','Hyderabad'),
(110,'Suresh Babu','9876543219','suresh@gmail.com','Vijayawada');
```

Output

Action Output

#	Time	Action	Message
2	22:26:57	USE Bank_Views_DB	0 row(s) affected
3	22:27:29	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	0 row(s) affected
4	22:27:59	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARC...	0 row(s) affected
5	22:28:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
6	22:28:48	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected
7	22:32:18	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101,'Harshika','9876543...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

SQL Views

```
(106,'Anirudh','9876543215','anirudh@gmail.com','Delhi'),
(107,'Goutham','9876543216','goutham@gmail.com','Mumbai'),
(108,'Zaizai','9876543217','zaizai@gmail.com','Pune'),
(109,'Manu','9876543218','manu@gmail.com','Hyderabad'),
(110,'Suresh Babu','9876543219','suresh@gmail.com','Vijayawada');
-- Step 7: Insert Account Data
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001,101,'Savings',50000,'Hyderabad'),
(10002,102,'Savings',75000,'Vijayawada'),
(10003,103,'Current',120000,'Bangalore'),
(10004,104,'Savings',45000,'Chennai'),
(10005,105,'Current',90000,'Hyderabad'),
(10006,106,'Savings',65000,'Delhi'),
(10007,107,'Current',150000,'Mumbai'),
(10008,108,'Savings',35000,'Pune'),
(10009,109,'Savings',85000,'Hyderabad'),
(10010,110,'Current',110000,'Vijayawada');
```

Output

Action Output

#	Time	Action	Message
3	22:27:29	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	0 row(s) affected
4	22:27:59	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARC...	0 row(s) affected
5	22:28:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
6	22:28:48	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected
7	22:32:18	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101,'Harshika','9876543...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	22:32:43	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001,101,'S...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

SQL Views

```
(10007,107,'Current',150000,'Mumbai'),
(10008,108,'Savings',35000,'Pune'),
(10009,109,'Savings',85000,'Hyderabad'),
(10010,110,'Current',110000,'Vijayawada');
-- Step 8: Insert Transaction Data
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001,'DEPOSIT',10000),
(10001,'WITHDRAW',5000),
(10002,'DEPOSIT',15000),
(10003,'WITHDRAW',20000),
(10004,'DEPOSIT',5000),
(10005,'WITHDRAW',10000),
(10006,'DEPOSIT',12000),
(10007,'DEPOSIT',25000),
(10008,'WITHDRAW',5000),
(10009,'DEPOSIT',20000),
(10010,'WITHDRAW',15000);
```

Output

Action Output

#	Time	Action	Message
4	22:27:59	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARC...	0 row(s) affected
5	22:28:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
6	22:28:48	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected
7	22:32:18	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101,'Harshika','9876543...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	22:32:43	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001,101,'S...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	22:33:17	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,'DEPOSIT',1000...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0

SQL Views

```
(10004,'DEPOSIT',5000),
(10005,'WITHDRAW',10000),
(10006,'DEPOSIT',12000),
(10007,'DEPOSIT',25000),
(10008,'WITHDRAW',5000),
(10009,'DEPOSIT',20000),
(10010,'WITHDRAW',15000);
-- Step 9: Insert Loan Data
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501,101,'Home Loan',5000000,7.5),
(502,102,'Education Loan',1000000,6.5),
(503,103,'Car Loan',800000,8.2),
(504,104,'Personal Loan',500000,10.5),
(505,105,'Home Loan',4000000,7.2),
(506,106,'Car Loan',900000,8.5),
(507,107,'Business Loan',3000000,9.0),
(508,109,'Personal Loan',600000,10.0);
```

Output

Action Output

#	Time	Action	Message
5	22:28:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
6	22:28:48	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected
7	22:32:18	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101,'Harshika','9876543...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	22:32:43	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001,101,'S...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	22:33:17	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,'DEPOSIT',1000...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0
10	22:33:42	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501,101,'Ho...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0

Step 2 : DISPLAY TABLES

SQL Views

Limit to 1000 rows

```
87      (502,102,'Education Loan',1000000,6.5),
88      (503,103,'Car Loan',800000,8.2),
89      (504,104,'Personal Loan',500000,10.5),
90      (505,105,'Home Loan',4000000,7.2),
91      (506,106,'Car Loan',900000,8.5),
92      (507,107,'Business Loan',3000000,9.0),
93      (508,109,'Personal Loan',600000,10.0);
94      -- Step 10: Display Original Tables
95 •    SELECT * FROM Customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Harshika	9876543210	harshika@gmail.com	Hyderabad
102	Ravinder	9876543211	ravi@gmail.com	Vijayawada
103	Santhvin	9876543212	santhvin@gmail.com	Bangalore
104	Srilatha	9876543213	srilatha@gmail.com	Chennai
105	Varshika	9876543214	varshika@gmail.com	erabad
106	Anirudh	9876543215	anirudh@gmail.com	Delhi
107	Goutham	9876543216	goutham@gmail.com	Mumbai
108	Zaizai	9876543217	zaizai@gmail.com	Pune
109	Manu	9876543218	manu@gmail.com	Hyderabad

Customer 1

Output

Action Output

#	Time	Action	Message
6	22:28:48	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected
7	22:32:18	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101,'Harshika','987654...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	22:32:43	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001,101,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	22:33:17	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,'DEPOSIT',100...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0
10	22:33:42	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501,101,'H...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
11	22:34:52	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned

SQL Views

Limit to 1000 rows

```
88      (503,103,'Car Loan',800000,8.2),
89      (504,104,'Personal Loan',500000,10.5),
90      (505,105,'Home Loan',4000000,7.2),
91      (506,106,'Car Loan',900000,8.5),
92      (507,107,'Business Loan',3000000,9.0),
93      (508,109,'Personal Loan',600000,10.0);
94      -- Step 10: Display Original Tables
95 •    SELECT * FROM Customer;
96 •    SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

Account 2

Output

Action Output

#	Time	Action	Message
7	22:32:18	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101,'Harshika','987654...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	22:32:43	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001,101,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	22:33:17	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,'DEPOSIT',100...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0
10	22:33:42	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501,101,'H...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
11	22:34:52	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	22:35:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned

SQL Views

Limit to 1000 rows

```
(504,104,'Personal Loan',500000,10.5),
(505,105,'Home Loan',4000000,7.2),
(506,106,'Car Loan',900000,8.5),
(507,107,'Business Loan',3000000,9.0),
(508,109,'Personal Loan',600000,10.0);
-- Step 10: Display Original Tables
SELECT * FROM Customer;
SELECT * FROM Account;
SELECT * FROM Bank_Transaction;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-13 22:33:17
2	10001	WITHDRAW	5000.00	2026-08-13 22:33:17
3	10002	DEPOSIT	15000.00	2026-08-13 22:33:17
4	10003	WITHDRAW	20000.00	2026-08-13 22:33:17
5	10004	DEPOSIT	5000.00	2026-08-13 22:33:17
6	10005	WITHDRAW	10000.00	2026-08-13 22:33:17
7	10006	DEPOSIT	12000.00	2026-08-13 22:33:17
8	10007	DEPOSIT	25000.00	2026-08-13 22:33:17
9	10008	WITHDRAW	5000.00	2026-08-13 22:33:17

Bank_Transaction 3

Output

Action Output

#	Time	Action	Message
8	22:32:43	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001,101,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	22:33:17	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,'DEPOSIT',100...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0
10	22:33:42	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501,101,'H...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
11	22:34:52	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	22:35:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	22:35:27	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned

SQL Views

Limit to 1000 rows

```
(505,105,'Home Loan',4000000,7.2),
(506,106,'Car Loan',900000,8.5),
(507,107,'Business Loan',3000000,9.0),
(508,109,'Personal Loan',600000,10.0);
-- Step 10: Display Original Tables
SELECT * FROM Customer;
SELECT * FROM Account;
SELECT * FROM Bank_Transaction;
SELECT * FROM Loan;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00
NULL	NULL	NULL	NULL	NULL

Loan 4

Output

Action Output

#	Time	Action	Message
9	22:33:17	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001,'DEPOSIT',100...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0
10	22:33:42	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501,101,'H...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
11	22:34:52	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	22:35:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	22:35:27	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
14	22:35:46	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned

Step 3 : SIMPLE VIEWS

SQL Views

```
85    VALUES
86    (501,101,'Home Loan',5000000,7.5),
87    (502,102,'Education Loan',1000000,6.5),
88    (503,103,'Car Loan',800000,8.2),
89    (504,104,'Personal Loan',500000,10.5),
90    (505,105,'Home Loan',4000000,7.2),
91    (506,106,'Car Loan',900000,8.5),
92    (507,107,'Business Loan',3000000,9.0),
93    (508,109,'Personal Loan',600000,10.0);
94    -- Step 10: Display Original Tables
95    SELECT * FROM Customer;
96    SELECT * FROM Account;
97    SELECT * FROM Bank_Transaction;
98    SELECT * FROM Loan;
99    -- Step 11: Create Customer View
100   CREATE VIEW Customer_View AS
101   SELECT * FROM Customer;
102
103   SELECT * FROM Customer_View;
```

Output

Action Output

#	Time	Action	Message
10	22:33:42	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501,101,'H...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
11	22:34:52	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	22:35:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	22:35:27	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
14	22:35:46	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned
15	22:36:33	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected

SQL Views

```
95    SELECT * FROM Customer;
96    SELECT * FROM Account;
97    SELECT * FROM Bank_Transaction;
98    SELECT * FROM Loan;
99    -- Step 11: Create Customer View
100   CREATE VIEW Customer_View AS
101   SELECT * FROM Customer;
102
103   SELECT * FROM Customer_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Harshika	9876543210	harshika@gmail.com	Hyderabad
102	Ravinder	9876543211	ravi@gmail.com	Vijayawada
103	Santhvin	9876543212	santhvin@gmail.com	Bangalore
104	Srilatha	9876543213	srilatha@gmail.com	Chennai
105	Varshika	9876543214	varshika@gmail.com	Hyderabad
106	Anirudh	9876543215	anirudh@gmail.com	Delhi
107	Goutham	9876543216	goutham@gmail.com	Mumbai
108	Zaizai	9876543217	zaizai@gmail.com	Pune
109	Manu	9876543218	manu@gmail	

manu@gmail.com

Customer_View 5

Output

Action Output

#	Time	Action	Message
11	22:34:52	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	22:35:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	22:35:27	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
14	22:35:46	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned
15	22:36:33	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
16	22:36:51	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned

SQL Views

```
91    (506,106,'Car Loan',900000,8.5),
92    (507,107,'Business Loan',3000000,9.0),
93    (508,109,'Personal Loan',600000,10.0);
94    -- Step 10: Display Original Tables
95    SELECT * FROM Customer;
96    SELECT * FROM Account;
97    SELECT * FROM Bank_Transaction;
98    SELECT * FROM Loan;
99    -- Step 11: Create Customer View
100   CREATE VIEW Customer_View AS
101   SELECT * FROM Customer;
102
103   SELECT * FROM Customer_View;
104   -- Step 12: Create Customer Basic View
105   CREATE VIEW Customer_Basic_View AS
106   SELECT Customer_ID, Customer_Name, City
107   FROM Customer;
108
109   SELECT * FROM Customer_Basic_View;
```

Output

Action Output

#	Time	Action	Message
12	22:35:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	22:35:27	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
14	22:35:46	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned
15	22:36:33	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
16	22:36:51	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	22:37:19	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected

SQL Views

```
101   SELECT * FROM Customer;
102
103   SELECT * FROM Customer_View;
104   -- Step 12: Create Customer Basic View
105   CREATE VIEW Customer_Basic_View AS
106   SELECT Customer_ID, Customer_Name, City
107   FROM Customer;
108
109   SELECT * FROM Customer_Basic_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City
101	Harshika	Hyderabad
102	Ravinder	Vijayawada
103	Santhvin	Bangalore
104	Srilatha	Chennai
105	Varshika	Hyderabad
106	Anirudh	Delhi
107	Goutham	Mumbai
108	Zaizai	Pune
109	Manu	Hyderabad

Customer_Basic_View 6

Output

Action Output

#	Time	Action	Message
13	22:35:27	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
14	22:35:46	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned
15	22:36:33	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
16	22:36:51	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	22:37:19	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected
18	22:37:35	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned

SQL Views

```
97  SELECT * FROM Bank_Transaction;
98  SELECT * FROM Loan;
99  -- Step 11: Create Customer View
100 CREATE VIEW Customer_View AS
101 SELECT * FROM Customer;
102
103 SELECT * FROM Customer_View;
104 -- Step 12: Create Customer Basic View
105 CREATE VIEW Customer_Basic_View AS
106 SELECT Customer_ID, Customer_Name, City
107 FROM Customer;
108
109 SELECT * FROM Customer_Basic_View;
110 -- Step 13: Create Account View
111 CREATE VIEW Account_View AS
112 SELECT Account_No, Account_Type, Balance, Branch
113 FROM Account;
114
115 SELECT * FROM Account_View;
```

Output

Action Output

#	Time	Action	Message
14	22:35:46	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned
15	22:36:33	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
16	22:36:51	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	22:37:19	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected
18	22:37:35	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	22:37:59	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM Account	0 row(s) affected

SQL Views

```
107     FROM Customer;
108
109 •   SELECT * FROM Customer_Basic_View;
110     -- Step 13: Create Account View
111 •   CREATE VIEW Account_View AS
112     SELECT Account_No, Account_Type, Balance, Branch
113     FROM Account;
114
115 •   SELECT * FROM Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Account_Type	Balance	Branch
10001	Savings	50000.00	Hyderabad
10002	Savings	75000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	45000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	65000.00	Delhi
10007	Current	150000.00	Mumbai
10008	Savings	35000.00	Pune
10009	Savings	85000.00	Hyderabad

Account_View 7

Output

Action Output

#	Time	Action	Message
15	22:36:33	CREATE VIEW Customer_View AS SELECT * FROM Customer	0 row(s) affected
16	22:36:51	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	22:37:19	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected
18	22:37:35	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	22:37:59	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM Account	0 row(s) affected
20	22:38:15	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned

SQL Views

```
103 • SELECT * FROM Customer_View;
104   -- Step 12: Create Customer Basic View
105 • CREATE VIEW Customer_Basic_View AS
106   SELECT Customer_ID, Customer_Name, City
107   FROM Customer;
108
109 • SELECT * FROM Customer_Basic_View;
110   -- Step 13: Create Account View
111 • CREATE VIEW Account_View AS
112   SELECT Account_No, Account_Type, Balance, Branch
113   FROM Account;
114
115 • SELECT * FROM Account_View;
116   -- Step 14: Create Savings Account View
117 • CREATE VIEW Savings_Account_View AS
118   SELECT * FROM Account
119   WHERE Account_Type = 'Savings';
120
121 • SELECT * FROM Savings_Account_View;
```

Output

Action Output

#	Time	Action	Message
16	22:36:51	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	22:37:19	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected
18	22:37:35	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	22:37:59	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM Account	0 row(s) affected
20	22:38:15	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
21	22:39:36	CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings'	0 row(s) affected

SQL Views

```
113   FROM Account;
114
115 • SELECT * FROM Account_View;
116   -- Step 14: Create Savings Account View
117 • CREATE VIEW Savings_Account_View AS
118   SELECT * FROM Account
119   WHERE Account_Type = 'Savings';
120
121 • SELECT * FROM Savings_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

Savings_Account_View 8

Output

Action Output

#	Time	Action	Message
17	22:37:19	CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer	0 row(s) affected
18	22:37:35	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	22:37:59	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM Account	0 row(s) affected
20	22:38:15	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
21	22:39:36	CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings'	0 row(s) affected
22	22:39:54	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned

SQL Views

```
109  SELECT * FROM Customer_Basic_View;
110  -- Step 13: Create Account View
111  CREATE VIEW Account_View AS
112  SELECT Account_No, Account_Type, Balance, Branch
113  FROM Account;
114
115  SELECT * FROM Account_View;
116  -- Step 14: Create Savings Account View
117  CREATE VIEW Savings_Account_View AS
118  SELECT * FROM Account
119  WHERE Account_Type = 'Savings';
120
121  SELECT * FROM Savings_Account_View;
122  -- Step 15: Create Current Account View
123  CREATE VIEW Current_Account_View AS
124  SELECT * FROM Account
125  WHERE Account_Type = 'Current';
126
127  SELECT * FROM Current_Account_View;
```

Output

Action Output

#	Time	Action	Message
18	22:37:35	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	22:37:59	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM Account	0 row(s) affected
20	22:38:15	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
21	22:39:36	CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings'	0 row(s) affected
22	22:39:54	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned
23	22:40:24	CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Current'	0 row(s) affected

SQL Views

```
119  WHERE Account_Type = 'Savings';
120
121  SELECT * FROM Savings_Account_View;
122  -- Step 15: Create Current Account View
123  CREATE VIEW Current_Account_View AS
124  SELECT * FROM Account
125  WHERE Account_Type = 'Current';
126
127  SELECT * FROM Current_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10007	107	Current	150000.00	Mumbai
10010	110	Current	110000.00	Vijayawada

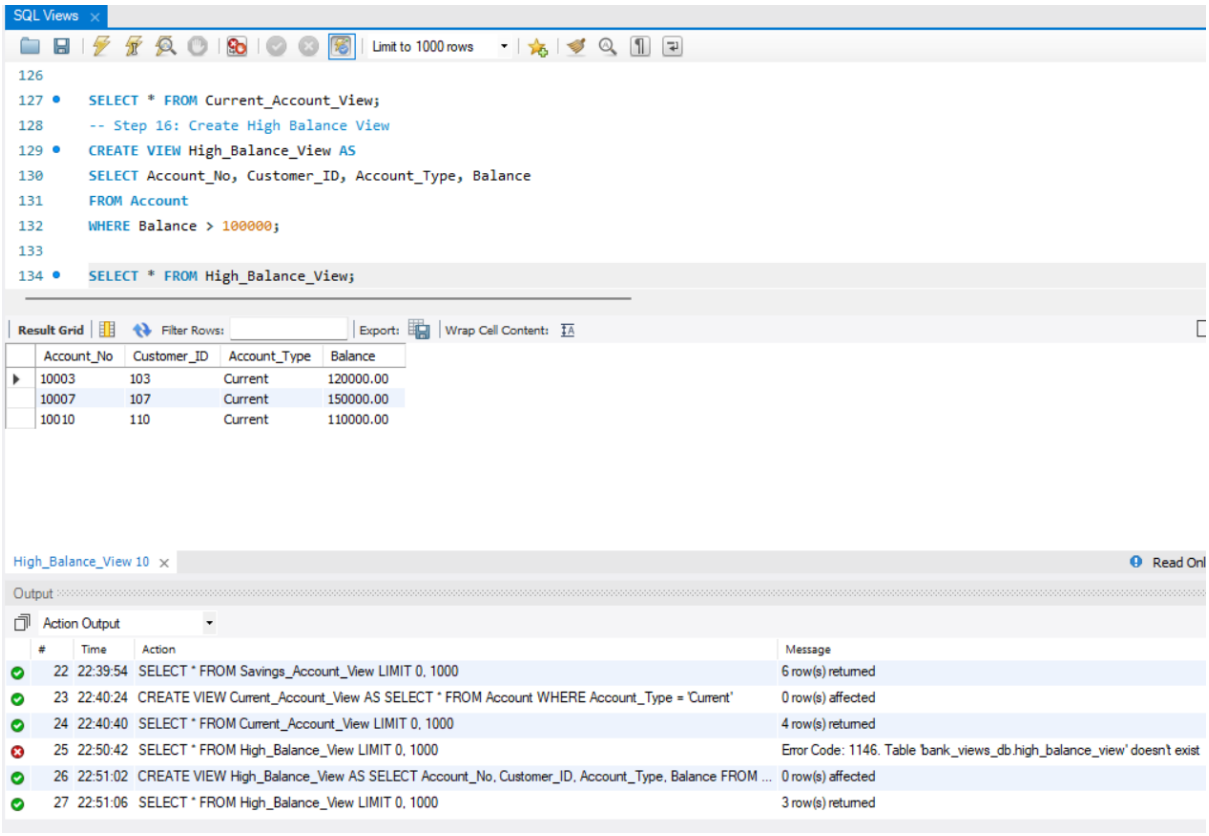
Current_Account_View 9

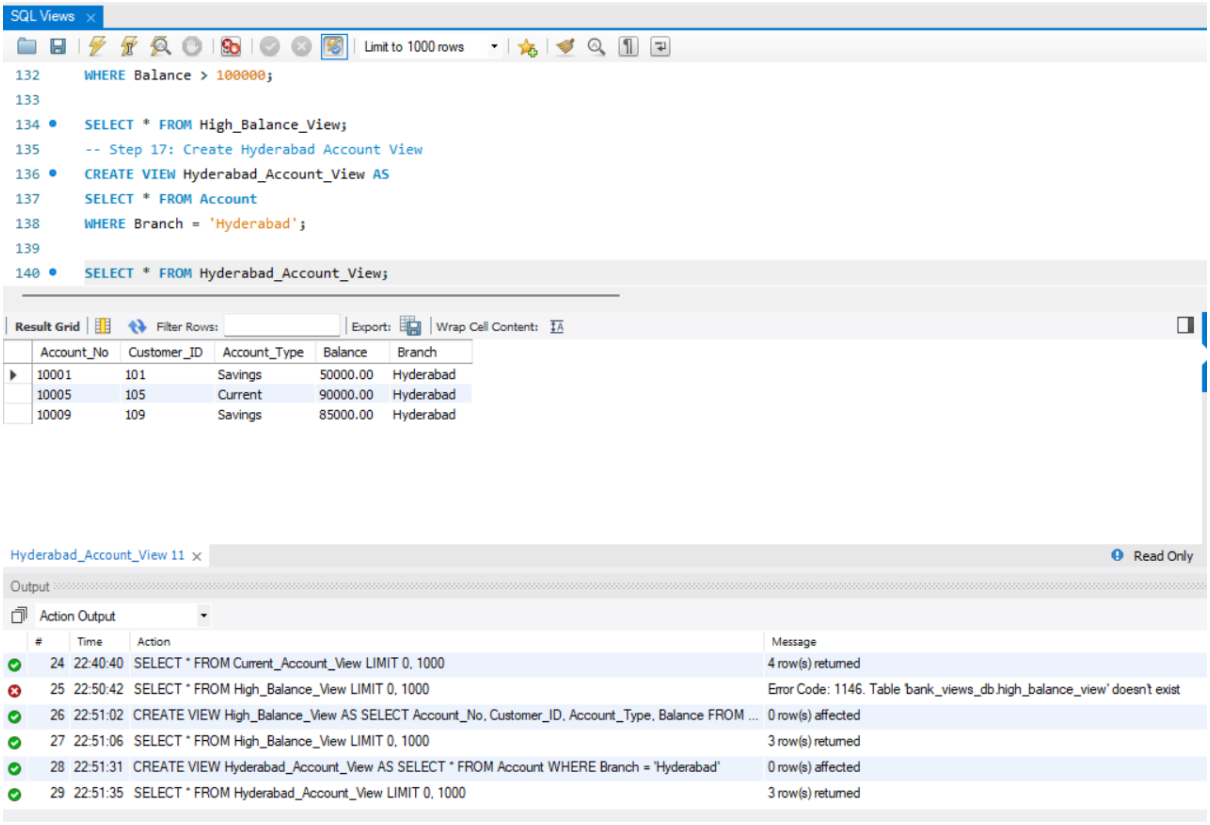
Output

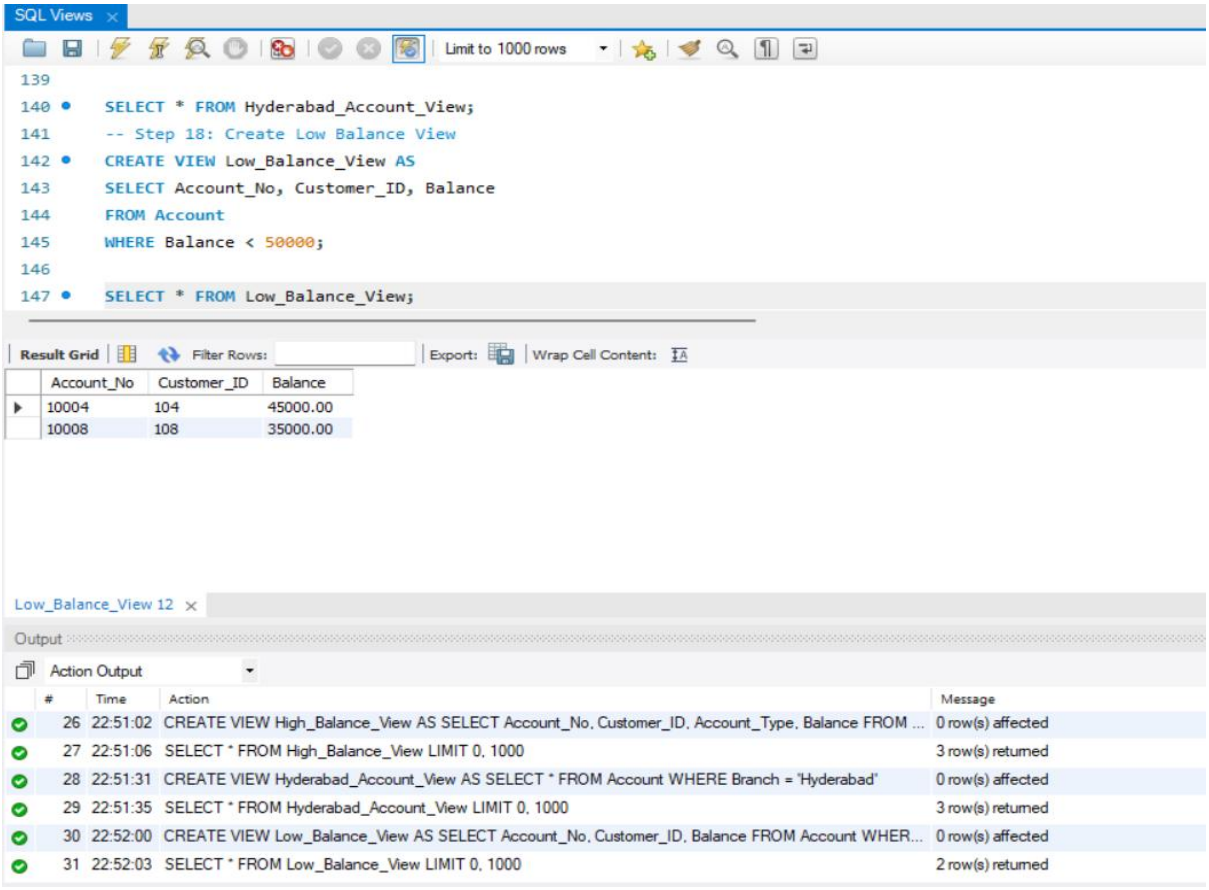
Action Output

#	Time	Action	Message
19	22:37:59	CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM Account	0 row(s) affected
20	22:38:15	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
21	22:39:36	CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings'	0 row(s) affected
22	22:39:54	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned
23	22:40:24	CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Current'	0 row(s) affected
24	22:40:40	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 row(s) returned

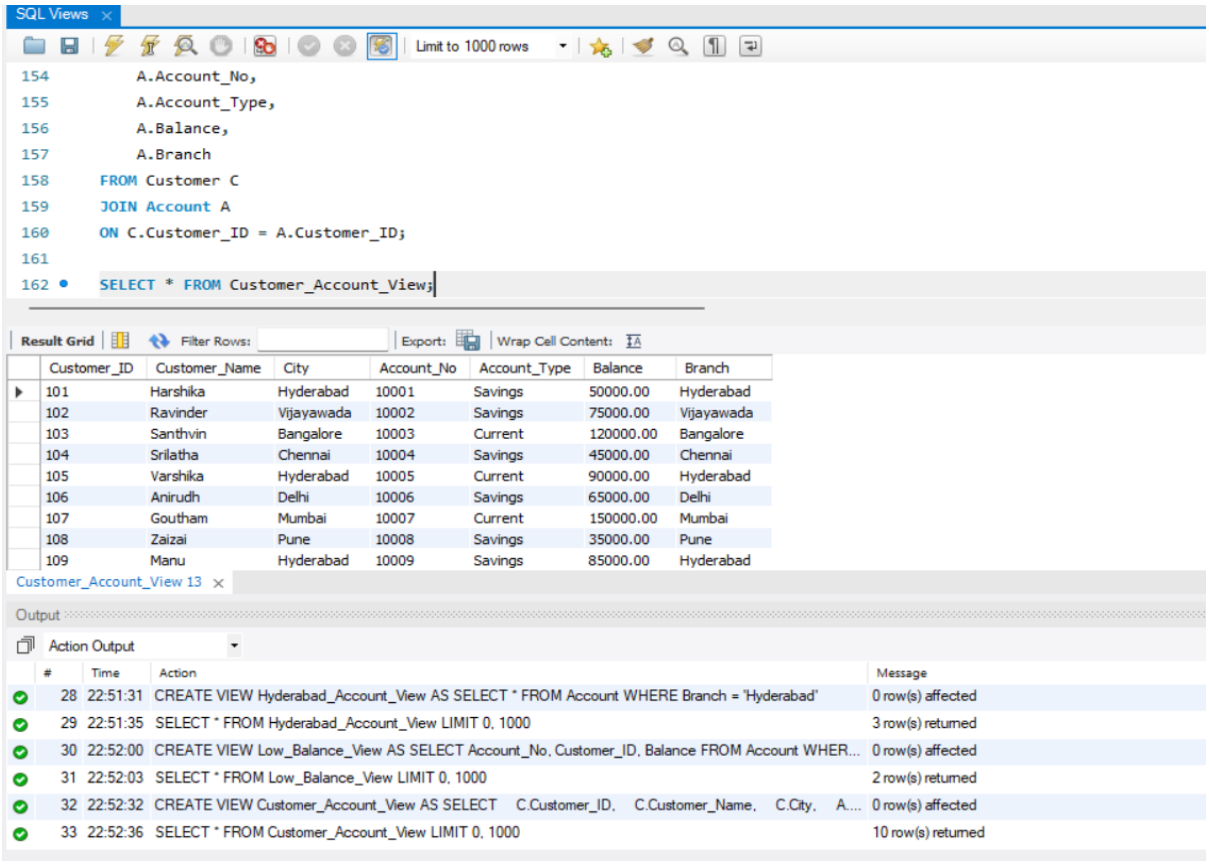
STEP 4 : CONDITIONAL VIEWS







STEP 5 : JOIN VIEWS



SQL Views

```
168         A.Account_No,
169         A.Account_Type,
170         A.Balance
171     FROM Customer C
172     JOIN Account A
173     ON C.Customer_ID = A.Customer_ID
174     WHERE A.Branch = 'Hyderabad';
175
176 •   SELECT * FROM Hyderabad_Customer_Accounts;
```

Result Grid

Customer_Name	City	Account_No	Account_Type	Balance
Harshika	Hyderabad	10001	Savings	50000.00
Varshika	Hyderabad	10005	Current	90000.00
Manu	Hyderabad	10009	Savings	85000.00

Hyderabad_Customer_Accounts...

Output

Action Output

#	Time	Action	Message
30	22:52:00	CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account WHER...	0 row(s) affected
31	22:52:03	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 row(s) returned
32	22:52:32	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A....	0 row(s) affected
33	22:52:36	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned
34	22:53:02	CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N...	0 row(s) affected
35	22:53:06	SELECT * FROM Hyderabad_Customer_Accounts LIMIT 0, 1000	3 row(s) returned

SQL Views

```
183         L.Loan_ID,
184         L.Loan_Type,
185         L.Loan_Amount,
186         L.Interest_Rate
187     FROM Customer C
188     JOIN Loan L
189     ON C.Customer_ID = L.Customer_ID;
190
191 •   SELECT * FROM Customer_Loan_View;
```

Result Grid

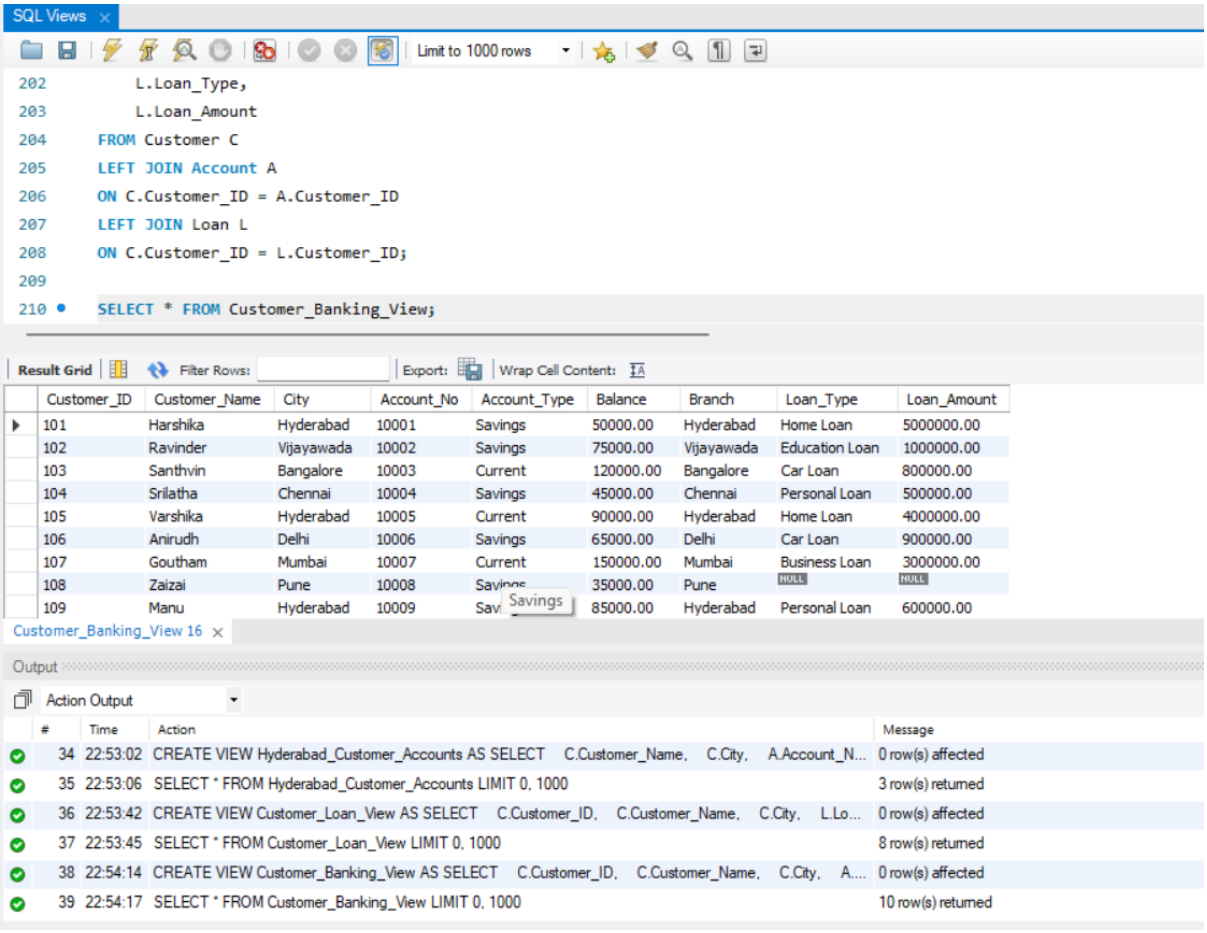
Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	Harshika	Hyderabad	501	Home Loan	5000000.00	7.50
102	Ravinder	Vijayawada	502	Education Loan	1000000.00	6.50
103	Santhvin	Bangalore	503	Car Loan	800000.00	8.20
104	Srilatha	Chennai	504	Personal Loan	500000.00	10.50
105	Varshika	Hyderabad	505	Home Loan	4000000.00	7.20
106	Anirudh	Delhi	506	Car Loan	900000.00	8.50
107	Goutham	Mumbai	507	Business Loan	3000000.00	9.00
109	Manu	Hyderabad	508	Personal Loan	600000.00	10.00

Customer_Loan_View 15

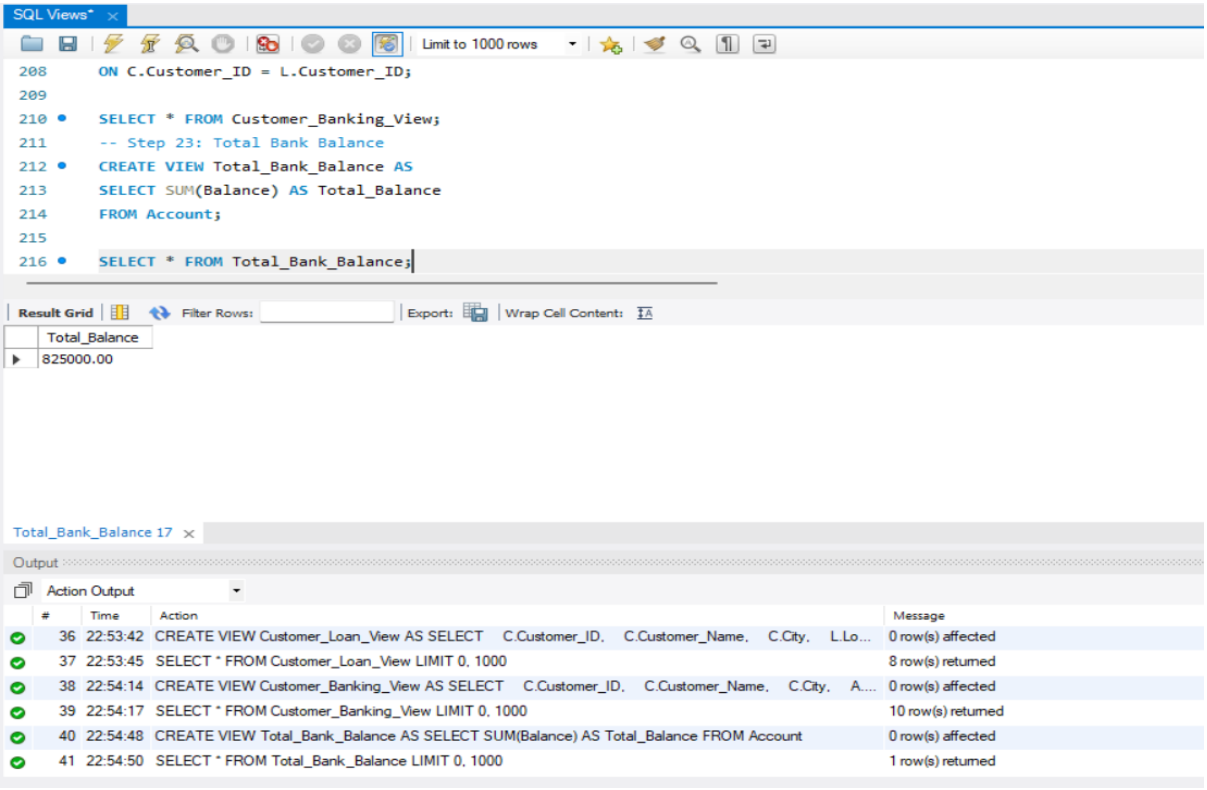
Output

Action Output

#	Time	Action	Message
32	22:52:32	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A....	0 row(s) affected
33	22:52:36	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned
34	22:53:02	CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N...	0 row(s) affected
35	22:53:06	SELECT * FROM Hyderabad_Customer_Accounts LIMIT 0, 1000	3 row(s) returned
36	22:53:42	CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, L.Lo...	0 row(s) affected
37	22:53:45	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	8 row(s) returned



STEP 6 : AGGREGATE VIEWS



SQL Views

```
214     FROM Account;
215
216 •   SELECT * FROM Total_Bank_Balance;
217     -- Step 24: Average Account Balance
218 •   CREATE VIEW Average_Account_Balance AS
219     SELECT AVG(Balance) AS Average_Balance
220     FROM Account;
221
222 •   SELECT * FROM Average_Account_Balance;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Average_Balance
82500.000000

Average_Account_Balance 18

Output

Action Output

#	Time	Action	Message
38	22:54:14	CREATE VIEW Customer_Banking_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A....	0 row(s) affected
39	22:54:17	SELECT * FROM Customer_Banking_View LIMIT 0, 1000	10 row(s) returned
40	22:54:48	CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account	0 row(s) affected
41	22:54:50	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned
42	22:55:14	CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Account	0 row(s) affected
43	22:55:17	SELECT * FROM Average_Account_Balance LIMIT 0, 1000	1 row(s) returned

SQL Views

```
220     FROM Account;
221
222 •   SELECT * FROM Average_Account_Balance;
223     -- Step 25: Maximum Account Balance
224 •   CREATE VIEW Maximum_Balance_View AS
225     SELECT MAX(Balance) AS Maximum_Balance
226     FROM Account;
227
228 •   SELECT * FROM Maximum_Balance_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Maximum_Balance
150000.00

Maximum_Balance_View 19

Output

Action Output

#	Time	Action	Message
40	22:54:48	CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account	0 row(s) affected
41	22:54:50	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned
42	22:55:14	CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Account	0 row(s) affected
43	22:55:17	SELECT * FROM Average_Account_Balance LIMIT 0, 1000	1 row(s) returned
44	22:55:56	CREATE VIEW Maximum_Balance_View AS SELECT MAX(Balance) AS Maximum_Balance FROM Account	0 row(s) affected
45	22:56:00	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned

SQL Views

```
226     FROM Account;
227
228 •   SELECT * FROM Maximum_Balance_View;
229     -- Step 26: Minimum Account Balance
230 •   CREATE VIEW Minimum_Balance_View AS
231     SELECT MIN(Balance) AS Minimum_Balance
232     FROM Account;
233
234 •   SELECT * FROM Minimum_Balance_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Minimum_Balance
35000.00

Minimum_Balance_View 20

Output

Action Output

#	Time	Action	Message
42	22:55:14	CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Account	0 row(s) affected
43	22:55:17	SELECT * FROM Average_Account_Balance LIMIT 0, 1000	1 row(s) returned
44	22:55:56	CREATE VIEW Maximum_Balance_View AS SELECT MAX(Balance) AS Maximum_Balance FROM Account	0 row(s) affected
45	22:56:00	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned
46	22:56:27	CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account	0 row(s) affected
47	22:56:30	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned

SQL Views

```
232     FROM Account;
233
234 •   SELECT * FROM Minimum_Balance_View;
235     -- Step 27: Total Number of Accounts
236 •   CREATE VIEW Account_Count_View AS
237     SELECT COUNT(*) AS Total_Accounts
238     FROM Account;
239
240 •   SELECT * FROM Account_Count_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Accounts
10

Account_Count_View 21

Output

Action Output

#	Time	Action	Message
44	22:55:56	CREATE VIEW Maximum_Balance_View AS SELECT MAX(Balance) AS Maximum_Balance FROM Account	0 row(s) affected
45	22:56:00	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned
46	22:56:27	CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account	0 row(s) affected
47	22:56:30	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned
48	22:57:02	CREATE VIEW Account_Count_View AS SELECT COUNT(*) AS Total_Accounts FROM Account	0 row(s) affected
49	22:57:05	SELECT * FROM Account_Count_View LIMIT 0, 1000	1 row(s) returned

STEP 7 : GROUP BY VIEWS

SQL Views*

```
239
240 •  SELECT * FROM Account_Count_View;
241    -- Step 28: Branch-wise Account Count
242 •  CREATE VIEW Branch_Account_Count AS
243    SELECT Branch, COUNT(*) AS Number_of_Accounts
244    FROM Account
245    GROUP BY Branch;
246
247 •  SELECT * FROM Branch_Account_Count;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2
Bangalore	1
Chennai	1
Delhi	1
Mumbai	1
Pune	1

Branch_Account_Count 22

Output

Action Output

#	Time	Action	Message
46	22:56:27	CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account	0 row(s) affected
47	22:56:30	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned
48	22:57:02	CREATE VIEW Account_Count_View AS SELECT COUNT(*) AS Total_Accounts FROM Account	0 row(s) affected
49	22:57:05	SELECT * FROM Account_Count_View LIMIT 0, 1000	1 row(s) returned
50	22:57:31	CREATE VIEW Branch_Account_Count AS SELECT Branch, COUNT(*) AS Number_of_Accounts FROM Acc...	0 row(s) affected
51	22:57:33	SELECT * FROM Branch_Account_Count LIMIT 0, 1000	7 row(s) returned

SQL Views*

```
246
247 •  SELECT * FROM Branch_Account_Count;
248    -- Step 29: Branch-wise Total Balance
249 •  CREATE VIEW Branch_Total_Balance AS
250    SELECT Branch, SUM(Balance) AS Total_Balance
251    FROM Account
252    GROUP BY Branch;
253
254 •  SELECT * FROM Branch_Total_Balance;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Chennai	45000.00
Delhi	65000.00
Mumbai	150000.00
Pune	35000.00

Branch_Total_Balance 23

Output

Action Output

#	Time	Action	Message
48	22:57:02	CREATE VIEW Account_Count_View AS SELECT COUNT(*) AS Total_Accounts FROM Account	0 row(s) affected
49	22:57:05	SELECT * FROM Account_Count_View LIMIT 0, 1000	1 row(s) returned
50	22:57:31	CREATE VIEW Branch_Account_Count AS SELECT Branch, COUNT(*) AS Number_of_Accounts FROM Acc...	0 row(s) affected
51	22:57:33	SELECT * FROM Branch_Account_Count LIMIT 0, 1000	7 row(s) returned
52	22:57:58	CREATE VIEW Branch_Total_Balance AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Accoun...	0 row(s) affected
53	22:58:00	SELECT * FROM Branch_Total_Balance LIMIT 0, 1000	7 row(s) returned

SQL Views

```
253
254 •  SELECT * FROM Branch_Total_Balance;
255    -- Step 30: Account Type Count
256 •  CREATE VIEW Account_Type_Count AS
257    SELECT Account_Type, COUNT(*) AS Number_of_Accounts
258    FROM Account
259    GROUP BY Account_Type;
260
261 •  SELECT * FROM Account_Type_Count;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_Type	Number_of_Accounts
Savings	6
Current	4

Account_Type_Count 24

Output

Action Output

#	Time	Action	Message
50	22:57:31	CREATE VIEW Branch_Account_Count AS SELECT Branch, COUNT(*) AS Number_of_Accounts FROM Acc...	0 row(s) affected
51	22:57:33	SELECT * FROM Branch_Account_Count LIMIT 0, 1000	7 row(s) returned
52	22:57:58	CREATE VIEW Branch_Total_Balance AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Accoun...	0 row(s) affected
53	22:58:00	SELECT * FROM Branch_Total_Balance LIMIT 0, 1000	7 row(s) returned
54	22:58:31	CREATE VIEW Account_Type_Count AS SELECT Account_Type, COUNT(*) AS Number_of_Accounts FRO...	0 row(s) affected
55	22:58:35	SELECT * FROM Account_Type_Count LIMIT 0, 1000	2 row(s) returned

SQL Views

```
260
261 •  SELECT * FROM Account_Type_Count;
262    -- Step 31: Account Type Total Balance
263 •  CREATE VIEW Account_Type_Balance AS
264    SELECT Account_Type, SUM(Balance) AS Total_Balance
265    FROM Account
266    GROUP BY Account_Type;
267
268 •  SELECT * FROM Account_Type_Balance;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_Type	Total_Balance
Savings	355000.00
Current	470000.00

Account_Type_Balance 25

Output

Action Output

#	Time	Action	Message
52	22:57:58	CREATE VIEW Branch_Total_Balance AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Accoun...	0 row(s) affected
53	22:58:00	SELECT * FROM Branch_Total_Balance LIMIT 0, 1000	7 row(s) returned
54	22:58:31	CREATE VIEW Account_Type_Count AS SELECT Account_Type, COUNT(*) AS Number_of_Accounts FRO...	0 row(s) affected
55	22:58:35	SELECT * FROM Account_Type_Count LIMIT 0, 1000	2 row(s) returned
56	22:59:05	CREATE VIEW Account_Type_Balance AS SELECT Account_Type, SUM(Balance) AS Total_Balance FRO...	0 row(s) affected
57	22:59:08	SELECT * FROM Account_Type_Balance LIMIT 0, 1000	2 row(s) returned

STEP 8 : HAVING VIEWS

SQL Views

```
268 •  SELECT * FROM Account_Type_Balance;
269    -- Step 32: Branches with More Than One Account
270 •  CREATE VIEW Multiple_Account_Branches AS
271    SELECT Branch, COUNT(*) AS Number_of_Accounts
272    FROM Account
273    GROUP BY Branch
274    HAVING COUNT(*) > 1;
275
276 •  SELECT * FROM Multiple_Account_Branches;
```

Result Grid

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2

Multiple_Account_Branches 26

Output

Action Output

#	Time	Action	Message
54	22:58:31	CREATE VIEW Account_Type_Count AS SELECT Account_Type, COUNT(*) AS Number_of_Accounts FRO...	0 row(s) affected
55	22:58:35	SELECT * FROM Account_Type_Count LIMIT 0, 1000	2 row(s) returned
56	22:59:05	CREATE VIEW Account_Type_Balance AS SELECT Account_Type, SUM(Balance) AS Total_Balance FRO...	0 row(s) affected
57	22:59:08	SELECT * FROM Account_Type_Balance LIMIT 0, 1000	2 row(s) returned
58	22:59:40	CREATE VIEW Multiple_Account_Branches AS SELECT Branch, COUNT(*) AS Number_of_Accounts FROM ...	0 row(s) affected
59	22:59:42	SELECT * FROM Multiple_Account_Branches LIMIT 0, 1000	2 row(s) returned

SQL Views*

```
276 •  SELECT * FROM Multiple_Account_Branches;
277    -- Step 33: Rich Branches
278 •  CREATE VIEW Rich_Branches AS
279    SELECT Branch, SUM(Balance) AS Total_Balance
280    FROM Account
281    GROUP BY Branch
282    HAVING SUM(Balance) > 100000;
283
284 •  SELECT * FROM Rich_Branches;
```

Result Grid

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Mumbai	150000.00

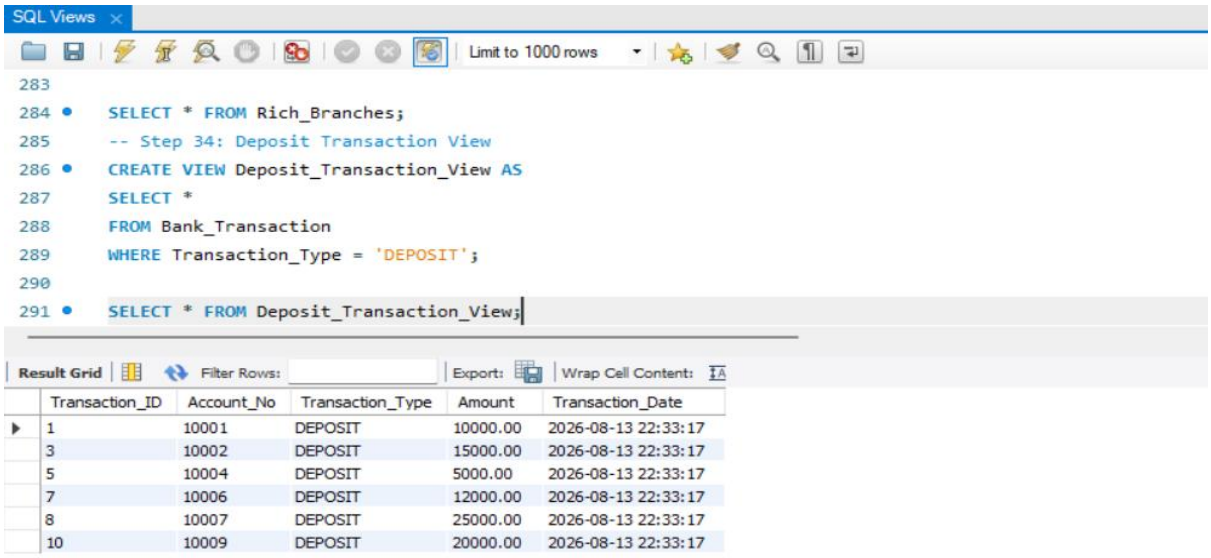
Rich_Branches 27

Output

Action Output

#	Time	Action	Message
56	22:59:05	CREATE VIEW Account_Type_Balance AS SELECT Account_Type, SUM(Balance) AS Total_Balance FRO...	0 row(s) affected
57	22:59:08	SELECT * FROM Account_Type_Balance LIMIT 0, 1000	2 row(s) returned
58	22:59:40	CREATE VIEW Multiple_Account_Branches AS SELECT Branch, COUNT(*) AS Number_of_Accounts FROM ...	0 row(s) affected
59	22:59:42	SELECT * FROM Multiple_Account_Branches LIMIT 0, 1000	2 row(s) returned
60	23:00:09	CREATE VIEW Rich_Branches AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Account GRO...	0 row(s) affected
61	23:00:12	SELECT * FROM Rich_Branches LIMIT 0, 1000	4 row(s) returned

STEP 9 : TRANSACTION VIEWS



```
283
284 •  SELECT * FROM Rich_Branches;
285    -- Step 34: Deposit Transaction View
286 •  CREATE VIEW Deposit_Transaction_View AS
287    SELECT *
288    FROM Bank_Transaction
289    WHERE Transaction_Type = 'DEPOSIT';
290
291 •  SELECT * FROM Deposit_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

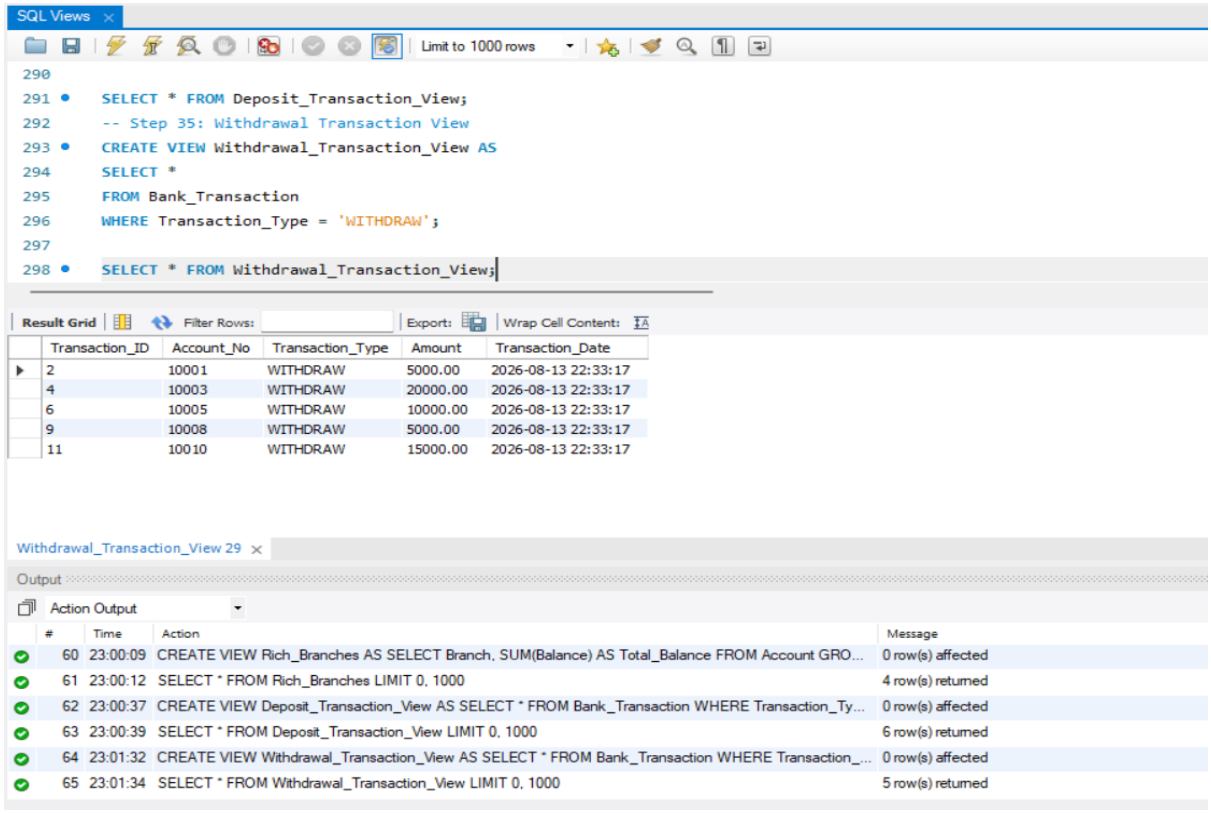
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-13 22:33:17
3	10002	DEPOSIT	15000.00	2026-08-13 22:33:17
5	10004	DEPOSIT	5000.00	2026-08-13 22:33:17
7	10006	DEPOSIT	12000.00	2026-08-13 22:33:17
8	10007	DEPOSIT	25000.00	2026-08-13 22:33:17
10	10009	DEPOSIT	20000.00	2026-08-13 22:33:17

Deposit_Transaction_View28

Output

Action Output

#	Time	Action	Message
58	22:59:40	CREATE VIEW Multiple_Account_Branches AS SELECT Branch, COUNT(*) AS Number_of_Accounts FROM ...	0 row(s) affected
59	22:59:42	SELECT * FROM Multiple_Account_Branches LIMIT 0, 1000	2 row(s) returned
60	23:00:09	CREATE VIEW Rich_Branches AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Account GRO...	0 row(s) affected
61	23:00:12	SELECT * FROM Rich_Branches LIMIT 0, 1000	4 row(s) returned
62	23:00:37	CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_Ty...	0 row(s) affected
63	23:00:39	SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000	6 row(s) returned



```
290
291 •  SELECT * FROM Deposit_Transaction_View;
292    -- Step 35: Withdrawal Transaction View
293 •  CREATE VIEW Withdrawal_Transaction_View AS
294    SELECT *
295    FROM Bank_Transaction
296    WHERE Transaction_Type = 'WITHDRAW';
297
298 •  SELECT * FROM Withdrawal_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
2	10001	WITHDRAW	5000.00	2026-08-13 22:33:17
4	10003	WITHDRAW	20000.00	2026-08-13 22:33:17
6	10005	WITHDRAW	10000.00	2026-08-13 22:33:17
9	10008	WITHDRAW	5000.00	2026-08-13 22:33:17
11	10010	WITHDRAW	15000.00	2026-08-13 22:33:17

Withdrawal_Transaction_View 29

Output

Action Output

#	Time	Action	Message
60	23:00:09	CREATE VIEW Rich_Branches AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Account GRO...	0 row(s) affected
61	23:00:12	SELECT * FROM Rich_Branches LIMIT 0, 1000	4 row(s) returned
62	23:00:37	CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_Ty...	0 row(s) affected
63	23:00:39	SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000	6 row(s) returned
64	23:01:32	CREATE VIEW Withdrawal_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_...	0 row(s) affected
65	23:01:34	SELECT * FROM Withdrawal_Transaction_View LIMIT 0, 1000	5 row(s) returned

SQL Views

```
307         T.Amount,
308         T.Transaction_Date
309     FROM Customer C
310     JOIN Account A
311     ON C.Customer_ID = A.Customer_ID
312     JOIN Bank_Transaction T
313     ON A.Account_No = T.Account_No;
314
315 •   SELECT * FROM Customer_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_Name	Account_No	Account_Type	Transaction_ID	Transaction_Type	Amount	Transaction_Date
Harshika	10001	Savings	1	DEPOSIT	10000.00	2026-08-13 22:33:17
Harshika	10001	Savings	2	WITHDRAW	5000.00	2026-08-13 22:33:17
Ravinder	10002	Savings	3	DEPOSIT	15000.00	2026-08-13 22:33:17
Santhvin	10003	Current	4	WITHDRAW	20000.00	2026-08-13 22:33:17
Srilatha	10004	Savings	5	DEPOSIT	5000.00	2026-08-13 22:33:17
Varshika	10005	Current	6	WITHDRAW	10000.00	2026-08-13 22:33:17
Anirudh	10006	Savings	7	DEPOSIT	12000.00	2026-08-13 22:33:17
Goutham	10007	Current	8	DEPOSIT	25000.00	2026-08-13 22:33:17
Zaizai	10008	Savings	9	WITHDRAW	5000.00	2026-08-13 22:33:17

Customer_Transaction_View 30

Output

Action Output

#	Time	Action	Message
62	23:00:37	CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_Ty...	0 row(s) affected
63	23:00:39	SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000	6 row(s) returned
64	23:01:32	CREATE VIEW Withdrawal_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_...	0 row(s) affected
65	23:01:34	SELECT * FROM Withdrawal_Transaction_View LIMIT 0, 1000	5 row(s) returned
66	23:02:00	CREATE VIEW Customer_Transaction_View AS SELECT C.Customer_Name, A.Account_No, A.Acco...	0 row(s) affected
67	23:02:03	SELECT * FROM Customer_Transaction_View LIMIT 0, 1000	11 row(s) returned

SQL Views

```
314
315 •   SELECT * FROM Customer_Transaction_View;
316     -- Step 37: High Value Transaction View
317 •   CREATE VIEW High_Value_Transaction_View AS
318     SELECT *
319     FROM Bank_Transaction
320     WHERE Amount > 10000;
321
322 •   SELECT * FROM High_Value_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
3	10002	DEPOSIT	15000.00	2026-08-13 22:33:17
4	10003	WITHDRAW	20000.00	2026-08-13 22:33:17
7	10006	DEPOSIT	12000.00	2026-08-13 22:33:17
8	10007	DEPOSIT	25000.00	2026-08-13 22:33:17
10	10009	DEPOSIT	20000.00	2026-08-13 22:33:17
11	10010	WITHDRAW	15000.00	2026-08-13 22:33:17

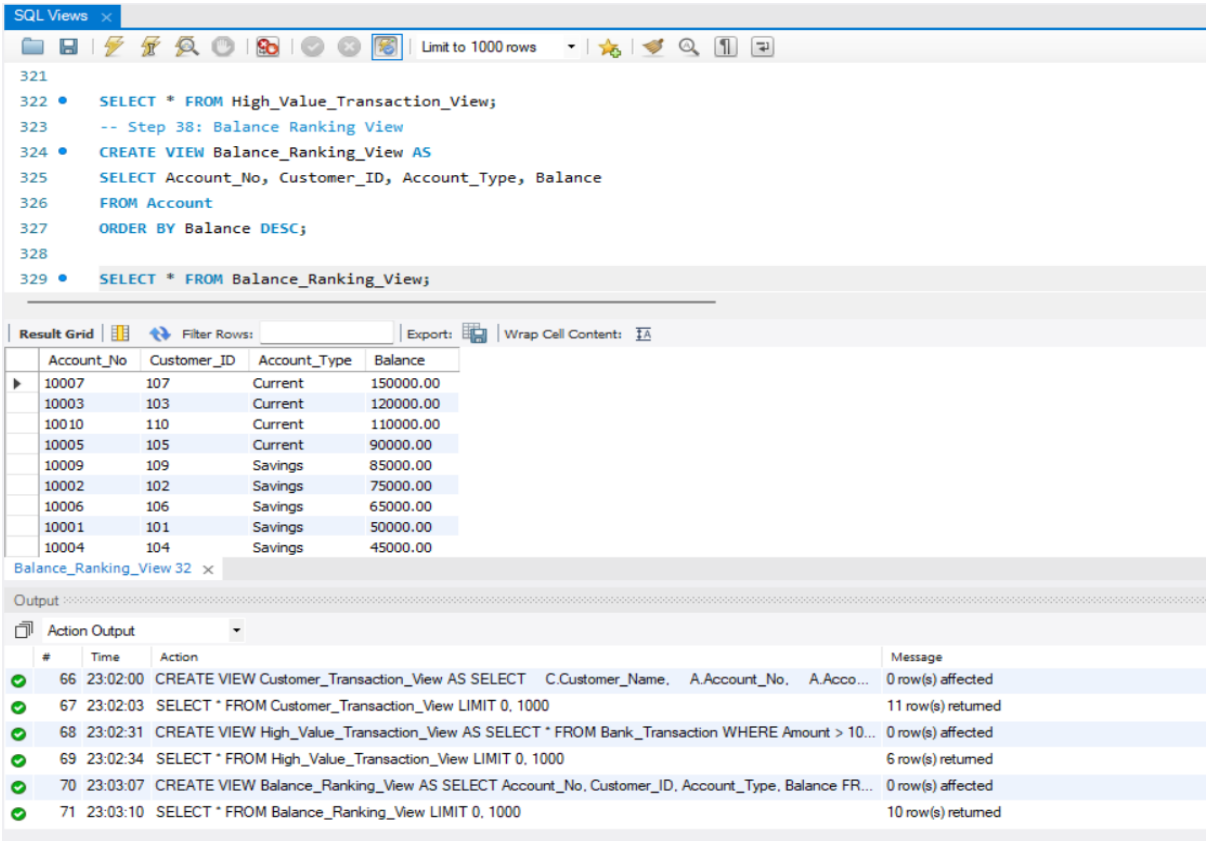
High_Value_Transaction_View 31

Output

Action Output

#	Time	Action	Message
64	23:01:32	CREATE VIEW Withdrawal_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_...	0 row(s) affected
65	23:01:34	SELECT * FROM Withdrawal_Transaction_View LIMIT 0, 1000	5 row(s) returned
66	23:02:00	CREATE VIEW Customer_Transaction_View AS SELECT C.Customer_Name, A.Account_No, A.Acco...	0 row(s) affected
67	23:02:03	SELECT * FROM Customer_Transaction_View LIMIT 0, 1000	11 row(s) returned
68	23:02:31	CREATE VIEW High_Value_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Amount > 10...	0 row(s) affected
69	23:02:34	SELECT * FROM High_Value_Transaction_View LIMIT 0, 1000	6 row(s) returned

STEP 10 : ORDER BY VIEWS



SQL Views

Limit to 1000 rows

```
321
322 •  SELECT * FROM High_Value_Transaction_View;
323    -- Step 38: Balance Ranking View
324 •  CREATE VIEW Balance_Ranking_View AS
325    SELECT Account_No, Customer_ID, Account_Type, Balance
326    FROM Account
327    ORDER BY Balance DESC;
328
329 •  SELECT * FROM Balance_Ranking_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

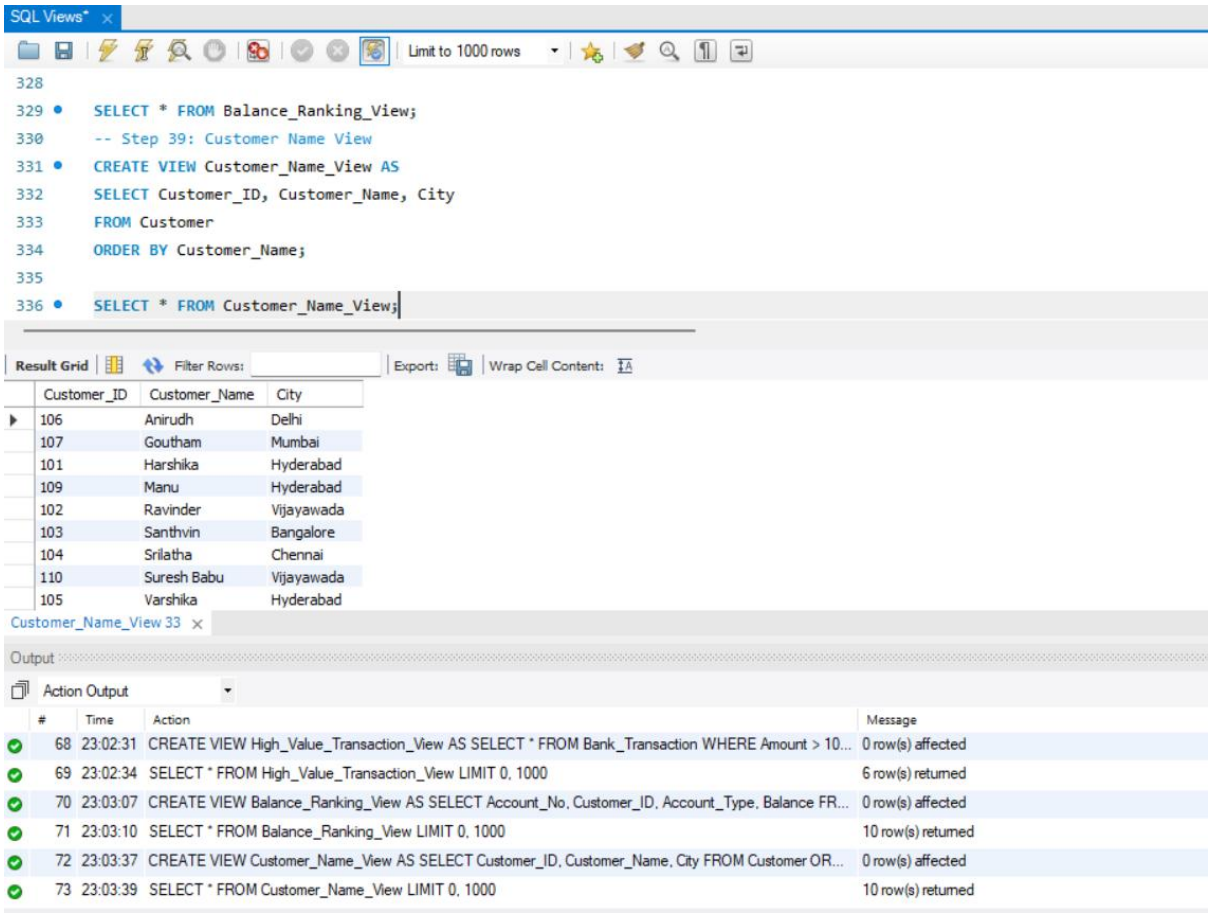
Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	150000.00
10003	103	Current	120000.00
10010	110	Current	110000.00
10005	105	Current	90000.00
10009	109	Savings	85000.00
10002	102	Savings	75000.00
10006	106	Savings	65000.00
10001	101	Savings	50000.00
10004	104	Savings	45000.00

Balance_Ranking_View 32

Output

Action Output

#	Time	Action	Message
66	23:02:00	CREATE VIEW Customer_Transaction_View AS SELECT C.Customer_Name, A.Account_No, A.Acco...	0 row(s) affected
67	23:02:03	SELECT * FROM Customer_Transaction_View LIMIT 0, 1000	11 row(s) returned
68	23:02:31	CREATE VIEW High_Value_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Amount > 10...	0 row(s) affected
69	23:02:34	SELECT * FROM High_Value_Transaction_View LIMIT 0, 1000	6 row(s) returned
70	23:03:07	CREATE VIEW Balance_Ranking_View AS SELECT Account_No, Customer_ID, Account_Type, Balance FR...	0 row(s) affected
71	23:03:10	SELECT * FROM Balance_Ranking_View LIMIT 0, 1000	10 row(s) returned



SQL Views*

Limit to 1000 rows

```
328
329 •  SELECT * FROM Balance_Ranking_View;
330    -- Step 39: Customer Name View
331 •  CREATE VIEW Customer_Name_View AS
332    SELECT Customer_ID, Customer_Name, City
333    FROM Customer
334    ORDER BY Customer_Name;
335
336 •  SELECT * FROM Customer_Name_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City
106	Anirudh	Delhi
107	Goutham	Mumbai
101	Harshika	Hyderabad
109	Manu	Hyderabad
102	Ravinder	Vijayawada
103	Santhvin	Bangalore
104	Srilatha	Chennai
110	Suresh Babu	Vijayawada
105	Varshika	Hyderabad

Customer_Name_View 33

Output

Action Output

#	Time	Action	Message
68	23:02:31	CREATE VIEW High_Value_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Amount > 10...	0 row(s) affected
69	23:02:34	SELECT * FROM High_Value_Transaction_View LIMIT 0, 1000	6 row(s) returned
70	23:03:07	CREATE VIEW Balance_Ranking_View AS SELECT Account_No, Customer_ID, Account_Type, Balance FR...	0 row(s) affected
71	23:03:10	SELECT * FROM Balance_Ranking_View LIMIT 0, 1000	10 row(s) returned
72	23:03:37	CREATE VIEW Customer_Name_View AS SELECT Customer_ID, Customer_Name, City FROM Customer OR...	0 row(s) affected
73	23:03:39	SELECT * FROM Customer_Name_View LIMIT 0, 1000	10 row(s) returned

STEP 11 : LOAN VIEWS

SQL Views

```
340         Loan_ID,
341         Customer_ID,
342         Loan_Type,
343         Loan_Amount,
344         Interest_Rate,
345         (Loan_Amount * Interest_Rate / 100) AS Annual_Interest
346     FROM Loan;
347
348 •   SELECT * FROM Loan_Interest_View;
```

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
501	101	Home Loan	5000000.00	7.50	375000.00000000
502	102	Education Loan	1000000.00	6.50	65000.00000000
503	103	Car Loan	800000.00	8.20	65600.00000000
504	104	Personal Loan	500000.00	10.50	52500.00000000
505	105	Home Loan	4000000.00	7.20	288000.00000000
506	106	Car Loan	900000.00	8.50	76500.00000000
507	107	Business Loan	3000000.00	9.00	270000.00000000
508	109	Personal Loan	600000.00	10.00	60000.00000000

Loan_Interest_View 34

Output

Action Output

#	Time	Action	Message
70	23:03:07	CREATE VIEW Balance_Ranking_View AS SELECT Account_No, Customer_ID, Account_Type, Balance FR...	0 row(s) affected
71	23:03:10	SELECT * FROM Balance_Ranking_View LIMIT 0, 1000	10 row(s) returned
72	23:03:37	CREATE VIEW Customer_Name_View AS SELECT Customer_ID, Customer_Name, City FROM Customer OR...	0 row(s) affected
73	23:03:39	SELECT * FROM Customer_Name_View LIMIT 0, 1000	10 row(s) returned
74	23:04:12	CREATE VIEW Loan_Interest_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_Amoun...	0 row(s) affected
75	23:04:15	SELECT * FROM Loan_Interest_View LIMIT 0, 1000	8 row(s) returned

SQL Views

```
352         Loan_ID,
353         Customer_ID,
354         Loan_Type,
355         Loan_Amount,
356         Interest_Rate,
357         Loan_Amount + (Loan_Amount * Interest_Rate / 100) AS Total_Amount
358     FROM Loan;
359
360 •   SELECT * FROM Loan_Total_Amount_View;
```

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
501	101	Home Loan	5000000.00	7.50	5375000.00000000
502	102	Education Loan	1000000.00	6.50	1065000.00000000
503	103	Car Loan	800000.00	8.20	865600.00000000
504	104	Personal Loan	500000.00	10.50	552500.00000000
505	105	Home Loan	4000000.00	7.20	4288000.00000000
506	106	Car Loan	900000.00	8.50	976500.00000000
507	107	Business Loan	3000000.00	9.00	3270000.00000000
508	109	Personal Loan	600000.00	10.00	660000.00000000

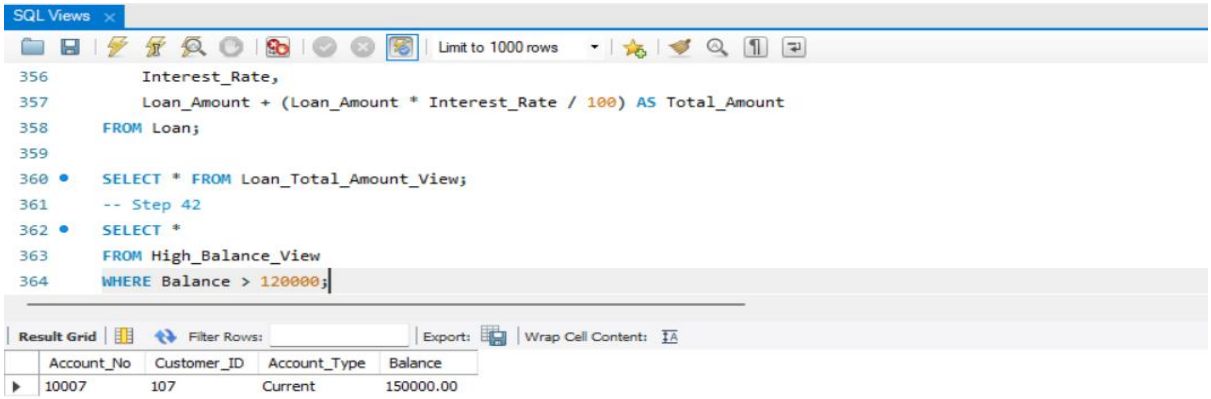
Loan_Total_Amount_View 35

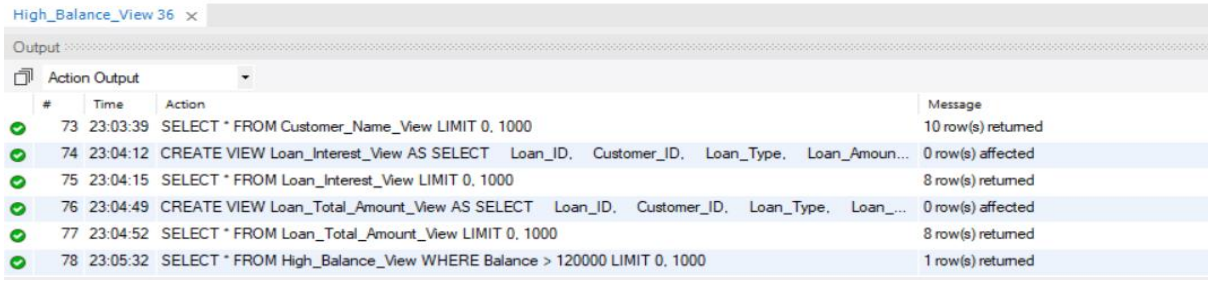
Output

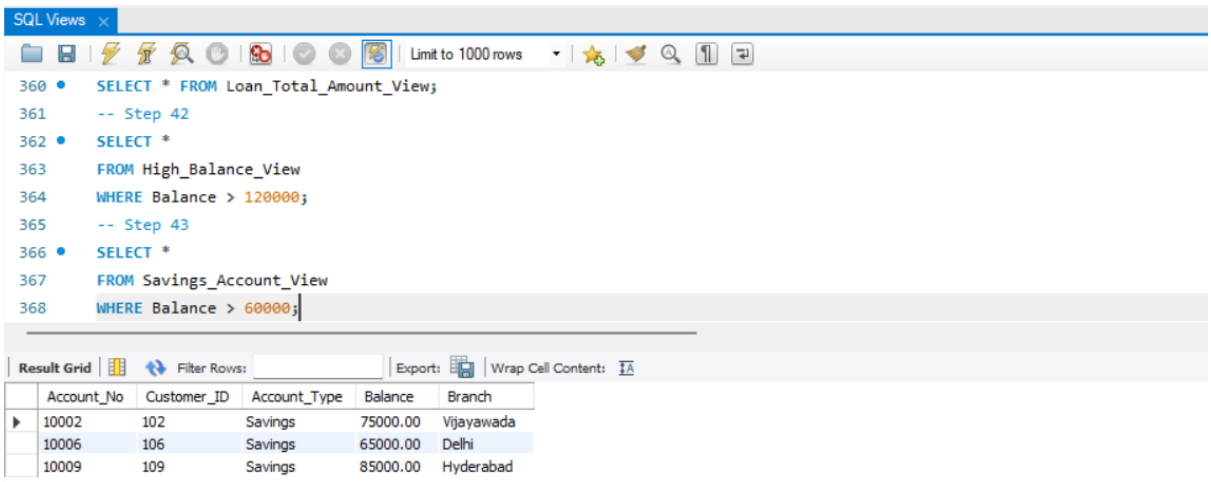
Action Output

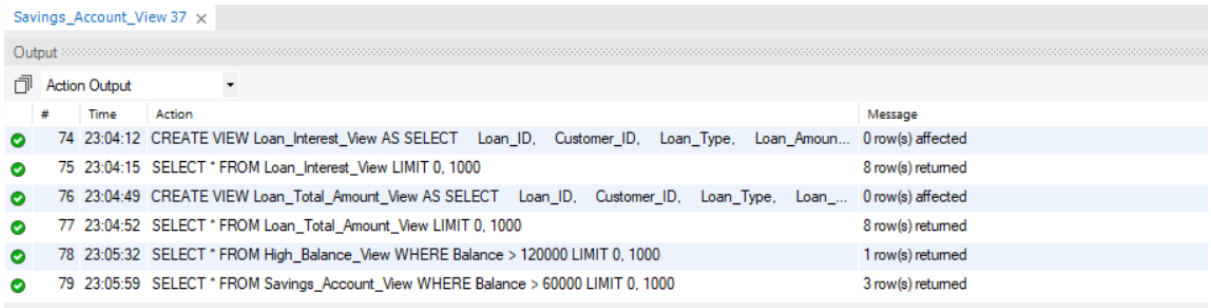
#	Time	Action	Message
72	23:03:37	CREATE VIEW Customer_Name_View AS SELECT Customer_ID, Customer_Name, City FROM Customer OR...	0 row(s) affected
73	23:03:39	SELECT * FROM Customer_Name_View LIMIT 0, 1000	10 row(s) returned
74	23:04:12	CREATE VIEW Loan_Interest_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_Amoun...	0 row(s) affected
75	23:04:15	SELECT * FROM Loan_Interest_View LIMIT 0, 1000	8 row(s) returned
76	23:04:49	CREATE VIEW Loan_Total_Amount_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_...	0 row(s) affected
77	23:04:52	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000	8 row(s) returned

STEP 12 : USING VIEWS WITH QUERIES









SQL Views

```
364     WHERE Balance > 120000;
365     -- Step 43
366 •   SELECT *
367     FROM Savings_Account_View
368     WHERE Balance > 60000;
369     -- Step 44
370 •   SELECT *
371     FROM Hyderabad_Account_View
372     WHERE Balance > 50000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10005	105	Current	90000.00	Hyderabad
10009	109	Savings	85000.00	Hyderabad

Hyderabad_Account_View 38

Output

Action Output

#	Time	Action	Message
75	23:04:15	SELECT * FROM Loan_Interest_View LIMIT 0, 1000	8 row(s) returned
76	23:04:49	CREATE VIEW Loan_Total_Amount_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_...	0 row(s) affected
77	23:04:52	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000	8 row(s) returned
78	23:05:32	SELECT * FROM High_Balance_View WHERE Balance > 120000 LIMIT 0, 1000	1 row(s) returned
79	23:05:59	SELECT * FROM Savings_Account_View WHERE Balance > 60000 LIMIT 0, 1000	3 row(s) returned
80	23:06:32	SELECT * FROM Hyderabad_Account_View WHERE Balance > 50000 LIMIT 0, 1000	2 row(s) returned

SQL Views

```
368     WHERE Balance > 60000;
369     -- Step 44
370 •   SELECT *
371     FROM Hyderabad_Account_View
372     WHERE Balance > 50000;
373     -- Step 45
374 •   SELECT *
375     FROM Customer_Basic_View
376     WHERE City = 'Hyderabad';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City
101	Harshika	Hyderabad
105	Varshika	Hyderabad
109	Manu	Hyderabad

Customer_Basic_View 39

Output

Action Output

#	Time	Action	Message
76	23:04:49	CREATE VIEW Loan_Total_Amount_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_...	0 row(s) affected
77	23:04:52	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000	8 row(s) returned
78	23:05:32	SELECT * FROM High_Balance_View WHERE Balance > 120000 LIMIT 0, 1000	1 row(s) returned
79	23:05:59	SELECT * FROM Savings_Account_View WHERE Balance > 60000 LIMIT 0, 1000	3 row(s) returned
80	23:06:32	SELECT * FROM Hyderabad_Account_View WHERE Balance > 50000 LIMIT 0, 1000	2 row(s) returned
81	23:06:56	SELECT * FROM Customer_Basic_View WHERE City = 'Hyderabad' LIMIT 0, 1000	3 row(s) returned

SQL Views

Limit to 1000 rows

```
372     WHERE Balance > 50000;
373     -- Step 45
374 •   SELECT *
375     FROM Customer_Basic_View
376     WHERE City = 'Hyderabad';
377     -- Step 46
378 •   SELECT *
379     FROM Customer_Account_View
380     ORDER BY Balance DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
107	Goutham	Mumbai	10007	Current	150000.00	Mumbai
103	Santhvin	Bangalore	10003	Current	120000.00	Bangalore
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada
105	Varshika	Hyderabad	10005	Current	90000.00	Hyderabad
109	Manu	Hyderabad	10009	Savings	85000.00	Hyderabad
102	Ravinder	Vijayawada	10002	Savings	75000.00	Vijayawada
106	Anirudh	Delhi	10006	Savings	65000.00	Delhi
101	Harshika	Hyderabad	10001	Savings	50000.00	Hyderabad
104	Srilatha	Chennai	10004	Savings	45000.00	Chennai

Customer_Account_View 40

Output

Action Output

#	Time	Action	Message
77	23:04:52	SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000	8 row(s) returned
78	23:05:32	SELECT * FROM High_Balance_View WHERE Balance > 120000 LIMIT 0, 1000	1 row(s) returned
79	23:05:59	SELECT * FROM Savings_Account_View WHERE Balance > 60000 LIMIT 0, 1000	3 row(s) returned
80	23:06:32	SELECT * FROM Hyderabad_Account_View WHERE Balance > 50000 LIMIT 0, 1000	2 row(s) returned
81	23:06:56	SELECT * FROM Customer_Basic_View WHERE City = 'Hyderabad' LIMIT 0, 1000	3 row(s) returned
82	23:07:41	SELECT * FROM Customer_Account_View ORDER BY Balance DESC LIMIT 0, 1000	10 row(s) returned

SQL Views

Limit to 1000 rows

```
376     WHERE City = 'Hyderabad';
377     -- Step 46
378 •   SELECT *
379     FROM Customer_Account_View
380     ORDER BY Balance DESC;
381     -- Step 47
382 •   SELECT *
383     FROM Customer_Loan_View
384     WHERE Loan_Amount > 1000000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	Harshika	Hyderabad	501	Home Loan	5000000.00	7.50
105	Varshika	Hyderabad	505	Home Loan	4000000.00	7.20
107	Goutham	Mumbai	507	Business Loan	3000000.00	9.00

Customer_Loan_View 41

Output

Action Output

#	Time	Action	Message
78	23:05:32	SELECT * FROM High_Balance_View WHERE Balance > 120000 LIMIT 0, 1000	1 row(s) returned
79	23:05:59	SELECT * FROM Savings_Account_View WHERE Balance > 60000 LIMIT 0, 1000	3 row(s) returned
80	23:06:32	SELECT * FROM Hyderabad_Account_View WHERE Balance > 50000 LIMIT 0, 1000	2 row(s) returned
81	23:06:56	SELECT * FROM Customer_Basic_View WHERE City = 'Hyderabad' LIMIT 0, 1000	3 row(s) returned
82	23:07:41	SELECT * FROM Customer_Account_View ORDER BY Balance DESC LIMIT 0, 1000	10 row(s) returned
83	23:08:15	SELECT * FROM Customer_Loan_View WHERE Loan_Amount > 1000000 LIMIT 0, 1000	3 row(s) returned

STEP 13 : UPDATE / INSERT / DELETE THROUGH VIEW

SQL Views*

```
384     WHERE Loan_Amount > 1000000;
385     -- Step 48: Update Account Balance Through View
386 •   UPDATE Account_View
387     SET Balance = 60000
388     WHERE Account_No = 10001;
389
390 •   SELECT *
391     FROM Account
392     WHERE Account_No = 10001;
```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 42

Output

Action Output

#	Time	Action	Message
80	23:06:32	SELECT * FROM Hyderabad_Account_View WHERE Balance > 50000 LIMIT 0, 1000	2 row(s) returned
81	23:06:56	SELECT * FROM Customer_Basic_View WHERE City = 'Hyderabad' LIMIT 0, 1000	3 row(s) returned
82	23:07:41	SELECT * FROM Customer_Account_View ORDER BY Balance DESC LIMIT 0, 1000	10 row(s) returned
83	23:08:15	SELECT * FROM Customer_Loan_View WHERE Loan_Amount > 1000000 LIMIT 0, 1000	3 row(s) returned
84	23:08:54	UPDATE Account_View SET Balance = 60000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
85	23:08:56	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned

SQL Views

```
399         Balance,
400         Branch
401     FROM Account;
402
403 •   INSERT INTO Simple_Account_View
404     VALUES
405     (10011,101,'Savings',55000,'Hyderabad');
406
407 •   SELECT * FROM Account;
```

Result Grid

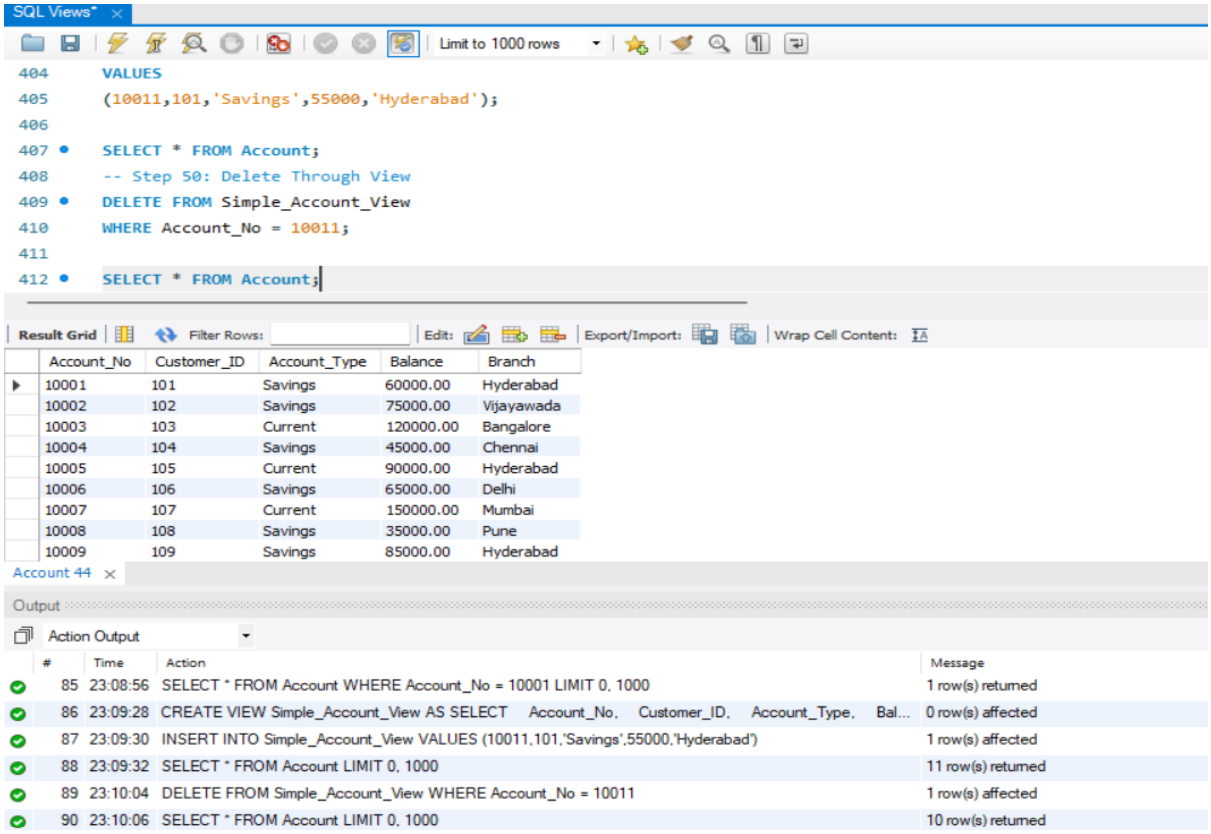
Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

Account 43

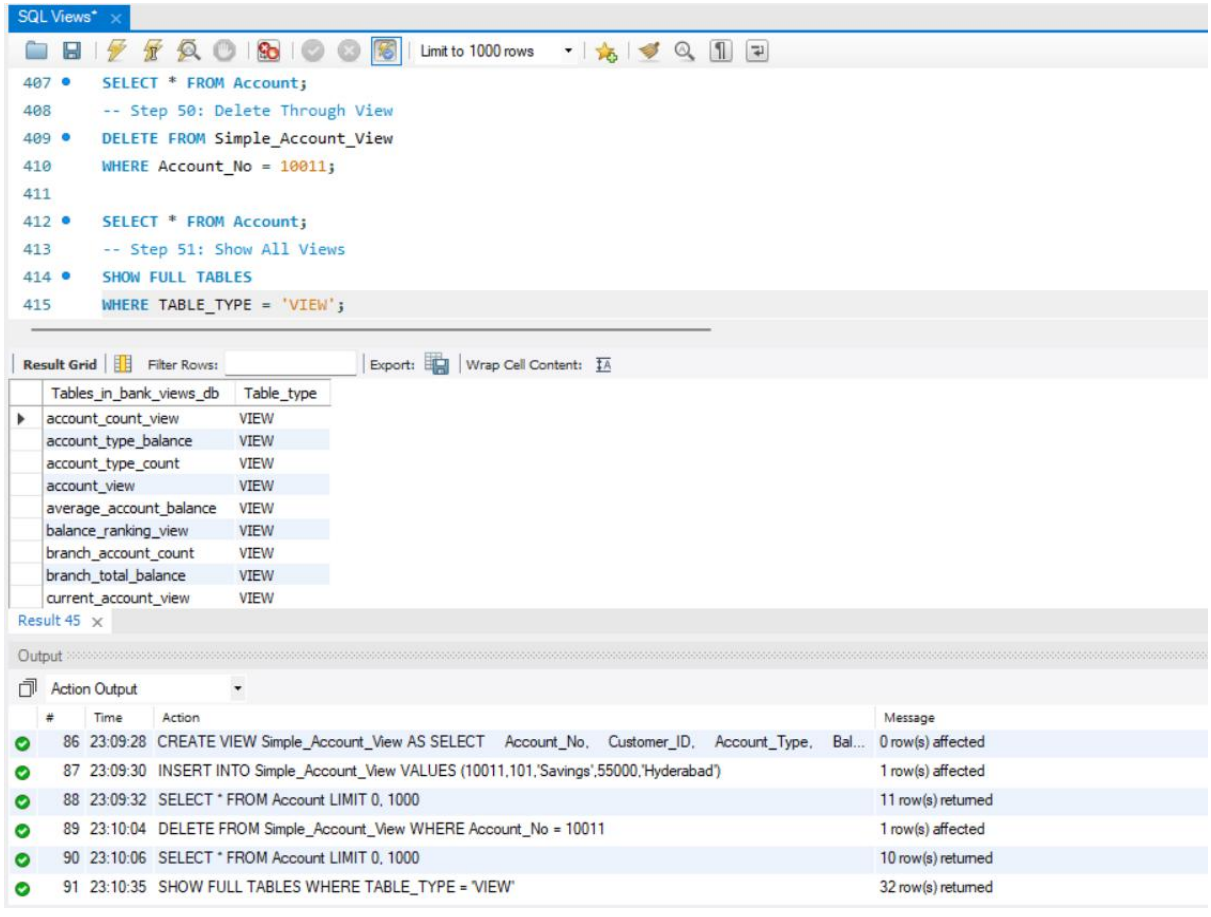
Output

Action Output

#	Time	Action	Message
83	23:08:15	SELECT * FROM Customer_Loan_View WHERE Loan_Amount > 1000000 LIMIT 0, 1000	3 row(s) returned
84	23:08:54	UPDATE Account_View SET Balance = 60000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
85	23:08:56	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned
86	23:09:28	CREATE VIEW Simple_Account_View AS SELECT Account_No, Customer_ID, Account_Type, Bal...	0 row(s) affected
87	23:09:30	INSERT INTO Simple_Account_View VALUES (10011,101,'Savings',55000,'Hyderabad')	1 row(s) affected
88	23:09:32	SELECT * FROM Account LIMIT 0, 1000	11 row(s) returned



STEP 14 : VIEW INFORMATION



SQL Views

Limit to 1000 rows

```
409 •  DELETE FROM Simple_Account_View
410    WHERE Account_No = 10011;
411
412 •  SELECT * FROM Account;
413    -- Step 51: Show All Views
414 •  SHOW FULL TABLES
415    WHERE TABLE_TYPE = 'VIEW';
416    -- Step 52: Show View Definition
417 •  SHOW CREATE VIEW Customer_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

View	Create View	character_set_client	collation_connection
customer_account_view	CREATE ALGORITHM=UNDEFINED DEFINER=`r...	utf8mb4	utf8mb4_0900_ai_ci

Result 46

Output

Action Output

#	Time	Action	Message
87	23:09:30	INSERT INTO Simple_Account_View VALUES (10011,101,'Savings',55000,'Hyderabad')	1 row(s) affected
88	23:09:32	SELECT * FROM Account LIMIT 0, 1000	11 row(s) returned
89	23:10:04	DELETE FROM Simple_Account_View WHERE Account_No = 10011	1 row(s) affected
90	23:10:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
91	23:10:35	SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW'	32 row(s) returned
92	23:11:02	SHOW CREATE VIEW Customer_Account_View	1 row(s) returned

SQL Views*

Limit to 1000 rows

```
411
412 •  SELECT * FROM Account;
413    -- Step 51: Show All Views
414 •  SHOW FULL TABLES
415    WHERE TABLE_TYPE = 'VIEW';
416    -- Step 52: Show View Definition
417 •  SHOW CREATE VIEW Customer_Account_View;
418    -- Step 53: Describe View
419 •  DESCRIBE Customer_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Customer_ID	int	NO		NULL	
Customer_Name	varchar(100)	NO		NULL	
City	varchar(50)	YES		NULL	
Account_No	int	NO		NULL	
Account_Type	varchar(20)	YES		NULL	
Balance	decimal(12,2)	YES		NULL	
Branch	varchar(50)	YES		NULL	

Result 47

Output

Action Output

#	Time	Action	Message
88	23:09:32	SELECT * FROM Account LIMIT 0, 1000	11 row(s) returned
89	23:10:04	DELETE FROM Simple_Account_View WHERE Account_No = 10011	1 row(s) affected
90	23:10:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
91	23:10:35	SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW'	32 row(s) returned
92	23:11:02	SHOW CREATE VIEW Customer_Account_View	1 row(s) returned
93	23:11:35	DESCRIBE Customer_Account_View	7 row(s) returned

STEP 15 : DROP VIEWS

SQL Views

```
403 •  INSERT INTO Simple_Account_View
404    VALUES
405    (10011,101,'Savings',55000,'Hyderabad');
406
407 •  SELECT * FROM Account;
408    -- Step 50: Delete Through View
409 •  DELETE FROM Simple_Account_View
410    WHERE Account_No = 10011;
411
412 •  SELECT * FROM Account;
413    -- Step 51: Show All Views
414 •  SHOW FULL TABLES
415    WHERE TABLE_TYPE = 'VIEW';
416    -- Step 52: Show View Definition
417 •  SHOW CREATE VIEW Customer_Account_View;
418    -- Step 53: Describe View
419 •  DESCRIBE Customer_Account_View;
420    -- Step 54: Drop One View
421 •  DROP VIEW Customer_View;
```

Output

Action Output

#	Time	Action	Message
89	23:10:04	DELETE FROM Simple_Account_View WHERE Account_No = 10011	1 row(s) affected
90	23:10:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
91	23:10:35	SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW'	32 row(s) returned
92	23:11:02	SHOW CREATE VIEW Customer_Account_View	1 row(s) returned
93	23:11:35	DESCRIBE Customer_Account_View	7 row(s) returned
94	23:12:06	DROP VIEW Customer_View	0 row(s) affected

SQL Views

```
408    -- Step 50: Delete Through View
409 •  DELETE FROM Simple_Account_View
410    WHERE Account_No = 10011;
411
412 •  SELECT * FROM Account;
413    -- Step 51: Show All Views
414 •  SHOW FULL TABLES
415    WHERE TABLE_TYPE = 'VIEW';
416    -- Step 52: Show View Definition
417 •  SHOW CREATE VIEW Customer_Account_View;
418    -- Step 53: Describe View
419 •  DESCRIBE Customer_Account_View;
420    -- Step 54: Drop One View
421 •  DROP VIEW Customer_View;
422    -- Step 55: Drop Multiple Views
423 •  DROP VIEW
424    Customer_Basic_View,
425    Savings_Account_View,
426    Current_Account_View;
```

Output

Action Output

#	Time	Action	Message
90	23:10:06	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
91	23:10:35	SHOW FULL TABLES WHERE TABLE_TYPE = 'VIEW'	32 row(s) returned
92	23:11:02	SHOW CREATE VIEW Customer_Account_View	1 row(s) returned
93	23:11:35	DESCRIBE Customer_Account_View	7 row(s) returned
94	23:12:06	DROP VIEW Customer_View	0 row(s) affected
95	23:12:38	DROP VIEW Customer_Basic_View, Savings_Account_View, Current_Account_View	0 row(s) affected